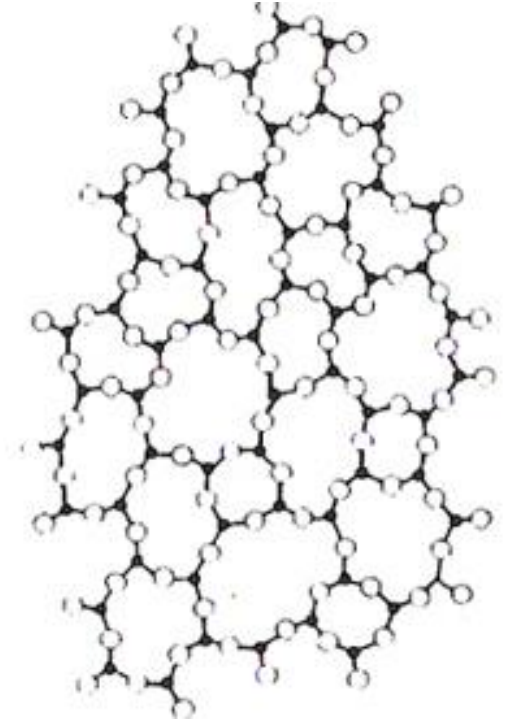


Materials Science

- **Materials science or materials engineering is an interdisciplinary field involving the properties of matter and its applications to various areas of science and engineering.**
- **This scientific field investigates the relationship between the structure of materials at atomic or molecular scales and their macroscopic properties.**
- **Materials science also deals with fundamental properties and characteristics of materials.**
- **The field encompasses the complete knowledge spectrum for materials ranging from the basic end (materials science) to the applied end (materials engineering).**

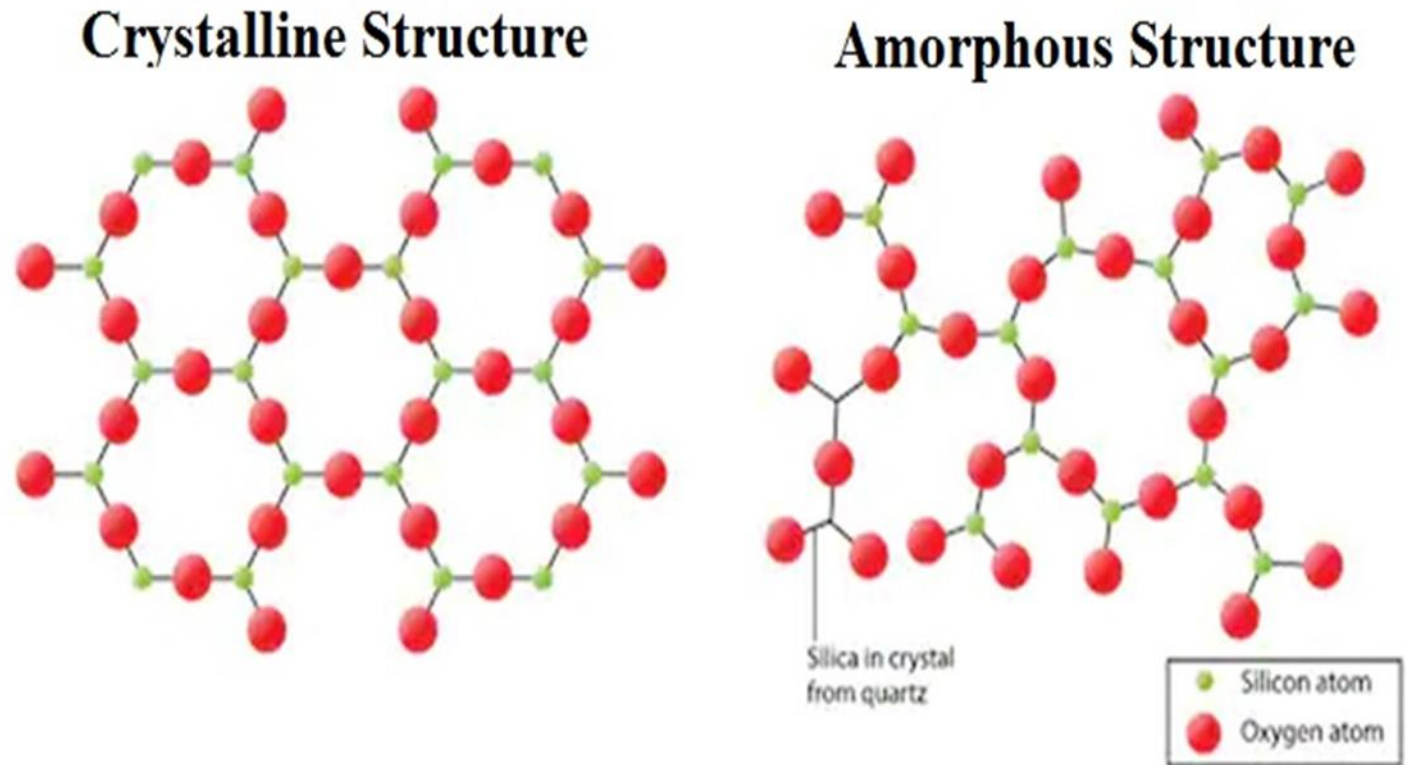


Materials Science

- **Materials science and engineering (MSE) is an interdisciplinary field concerned with inventing new materials and improving previously known materials by developing a deeper understanding of the microstructure-composition-synthesis-processing relationships.**
- **The term composition means the chemical make-up of a material.**
- **The term structure means a description of the arrangement of atoms, as seen at different levels of detail.**
- **Materials scientists and engineers not only deal with the development of materials, but also with the synthesis and processing of materials and manufacturing processes related to the production of components.**
- **The term “synthesis” refers to how materials are made from naturally occurring or man-made chemicals.**
- **The term “processing” means how materials are shaped into useful components.**
- **One of the most important functions of materials scientists and engineers is to establish the relationships between the properties of a material and its performance.**
- **In materials science, the emphasis is on the underlying relationships between the synthesis and processing, structure, and properties of materials.**
- **In materials engineering, the focus is on how to translate or transform materials into a useful device or structure.**

Types Of Materials

- **Materials can be divided into the following categories**
 - **Crystalline**
 - **Amorphous**

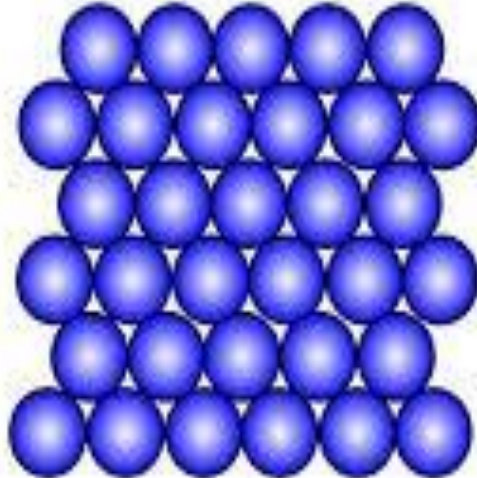


Crystalline Materials

- These are materials containing one or many crystals.
- In each crystal, atoms or ions show a long range periodic arrangement.
- All metals and alloys are crystalline materials.
- These include iron, steel, copper, brass, bronze, aluminium, uranium, thorium etc.

Crystalline Solid

Particles

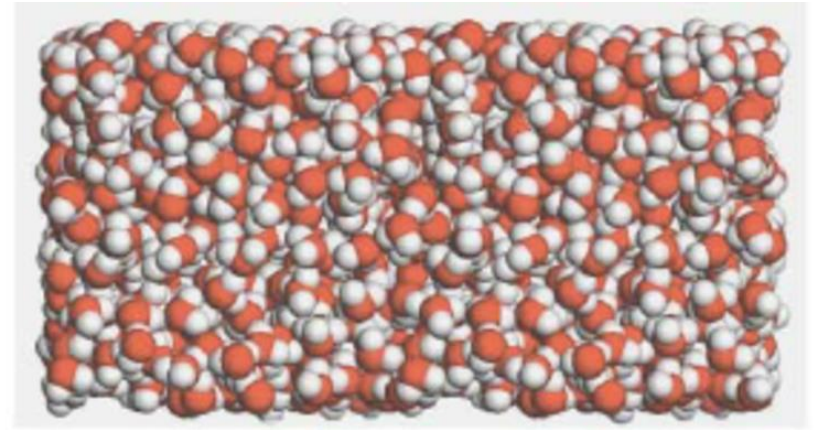


Example: Table Salt

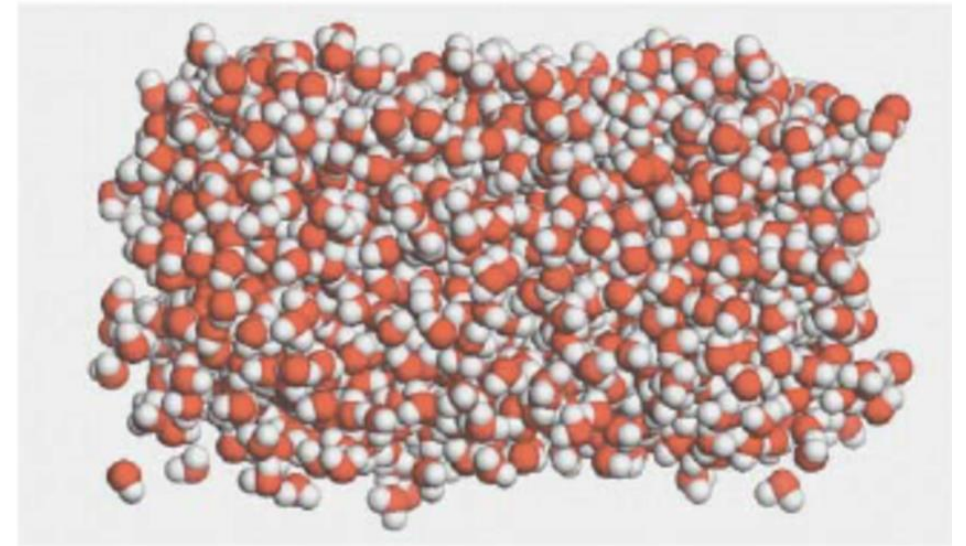


Amorphous Material

- The term amorphous refers to materials that do not have regular, periodic arrangement of atoms.
- Between solid and liquid.
- Glass is an amorphous material.

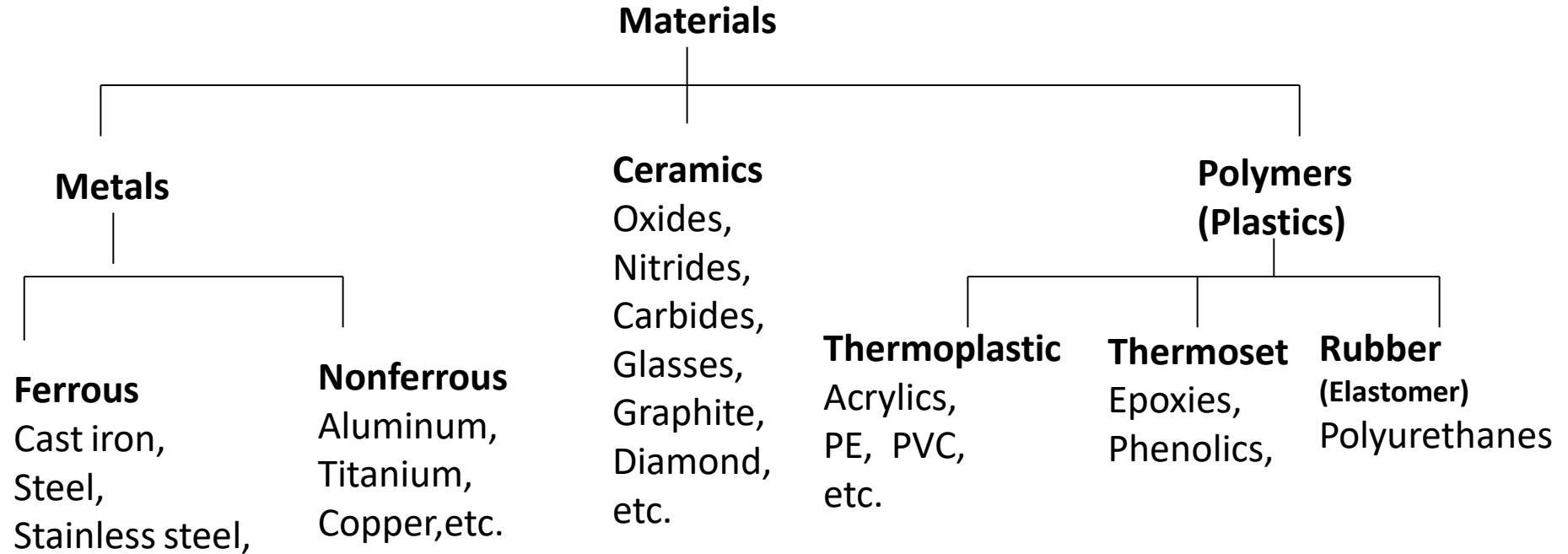


***(a)* glass**



***(b)* liquid**

Classification Of Engineering Materials



Composites-reinforced plastics, metal or ceramic matrix, laminates, others.

Semiconductors-Silicon, Germanium, Gallium phosphor, other.

Biomaterials-biocompatible materials, Co-Cr-Mo, Co-Ni-Mo metal alloys (for hip)

Difference Between Crystalline and *Amorphous Solids*

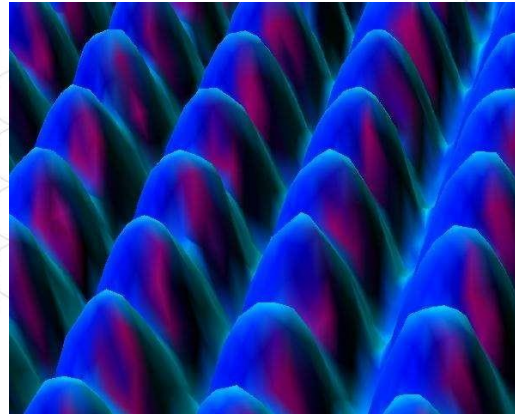
Properties	Crystalline solid	Amorphous solid
Geometrical shape	The crystals of every crystalline solid have a defined geometrical shape due to definite and orderly arrangement of particles in three-dimensional shape.	An amorphous solid does not have any orderly pattern of arrangement of particles and therefore, does not have any definite geometrical shape.
Melting point	The crystalline solids which directly change into liquid state have definite melting point.	Amorphous solids like glass do not have definite melting point.
Cleavage planes	Crystalline solids have definite cleavage planes which are inclined to one another at a particular angle for a given crystalline solid.	Amorphous solids do not have such well-defined cleavage planes.
Anisotropic and isotropic properties	As the orderly arranged particles in crystalline solids are different in different directions, these solids show anisotropic properties.	As in amorphous solids, as those of liquids and gases, the arrangement of particles is random, these solids show isotropic properties.
Symmetry	Crystalline solids have crystal symmetry.	Amorphous solids do not have symmetry.
Tensile strength	The tensile strength of crystalline solid is very high.	The tensile strength of amorphous solid is low.
Fracturing	The crystalline solids fracture in a ductile manner.	The amorphous solids fracture in a brittle manner.
Examples	Metals, <i>NaCl</i> , <i>ZnO</i> , <i>KCl</i> , <i>CsCl</i> , carbon etc.	Glass, plastics, polymers, rubbers, silica etc.

Crystal Lattice

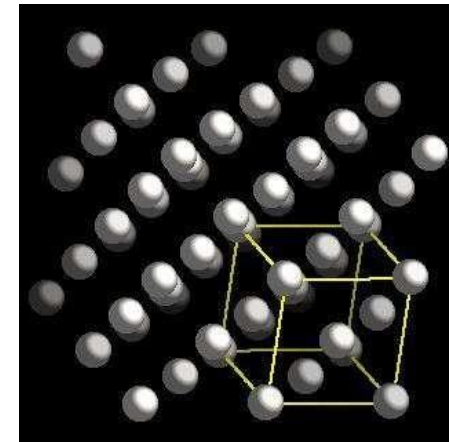
- What is crystal (space) lattice?
- In crystallography, only the geometrical properties of the crystal are of interest, therefore one replaces each atom by a geometrical point located at the equilibrium position of that atom.



Platinum



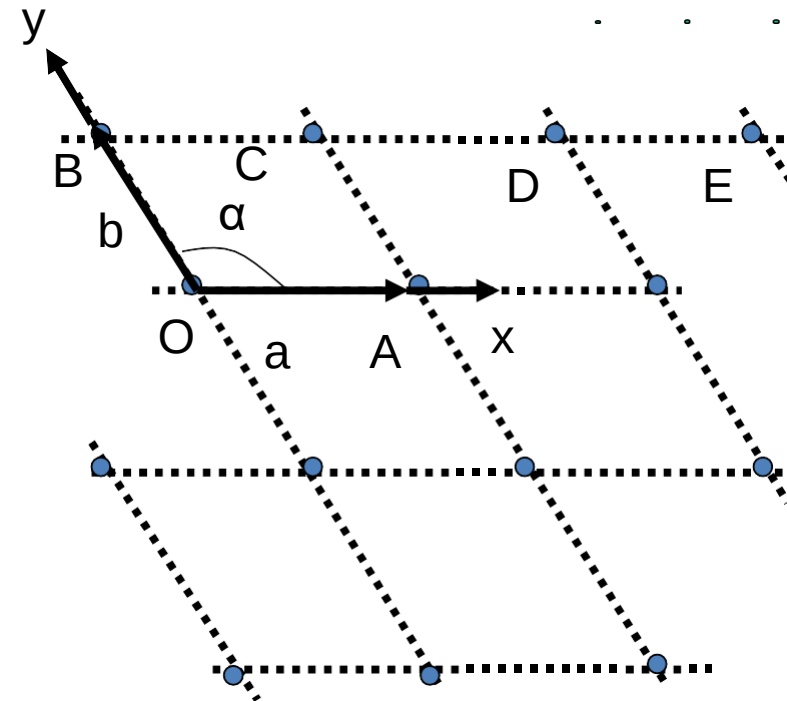
Platinum surface
(scanning tunneling microscope)



Crystal lattice and
structure of Platinum

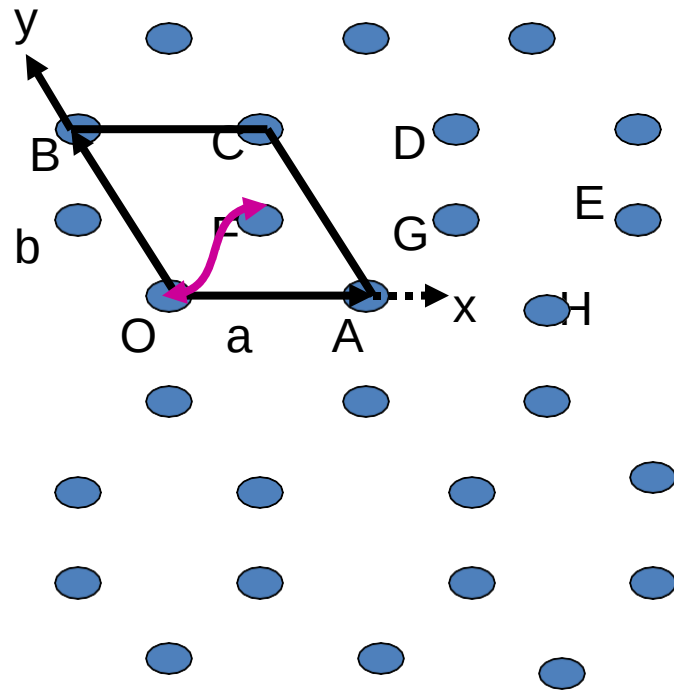
Crystal Lattice

- An infinite array of points in space.
- Each point has identical surroundings to all others.
- Arrays are arranged exactly in a periodic manner.

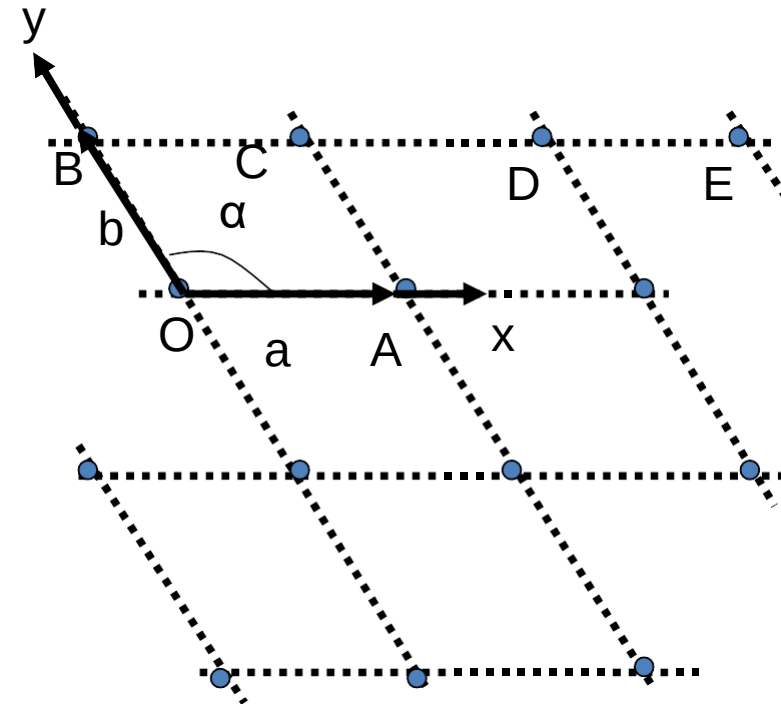


Basis [Motif]

- A group of atoms which describe crystal structure



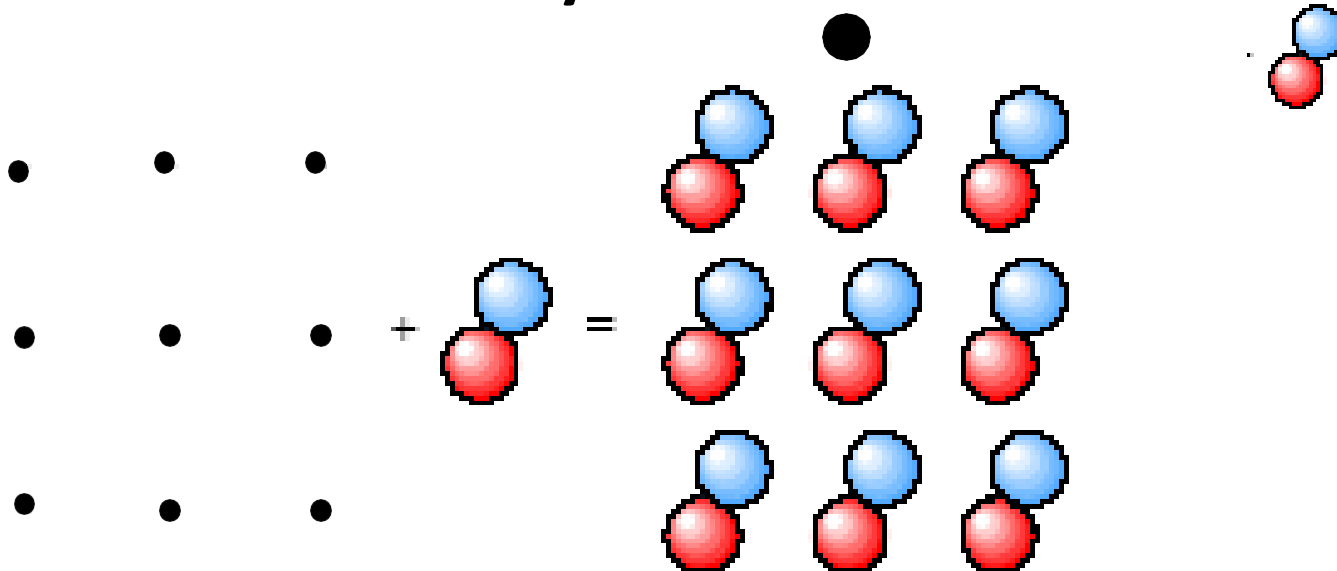
a) Situation of atoms at the corners of regular hexagons



b) Crystal lattice obtained by identifying all the atoms in (a)

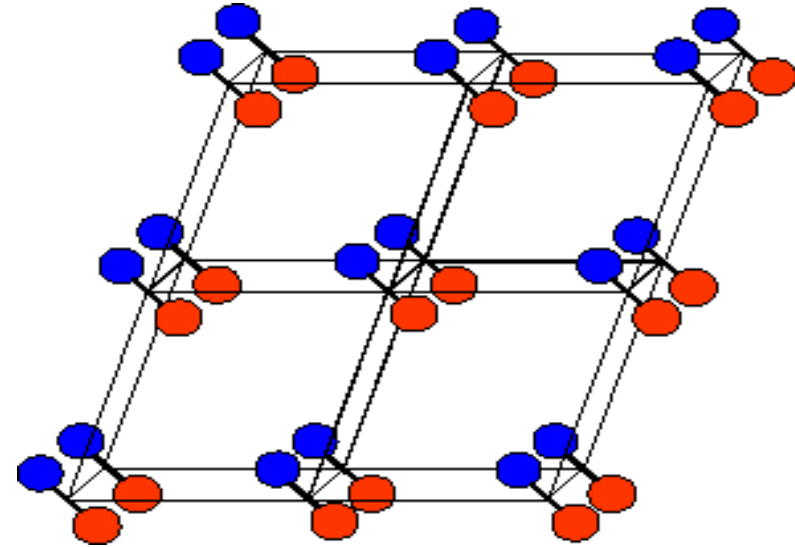
Crystal Structure

- When an atom or identical group of atoms or molecules which are called basis (motif) is attached to every lattice point, we obtain a crystal structure.
- **Crystal Structure = Crystal Lattice + Basis**



Crystal Structure

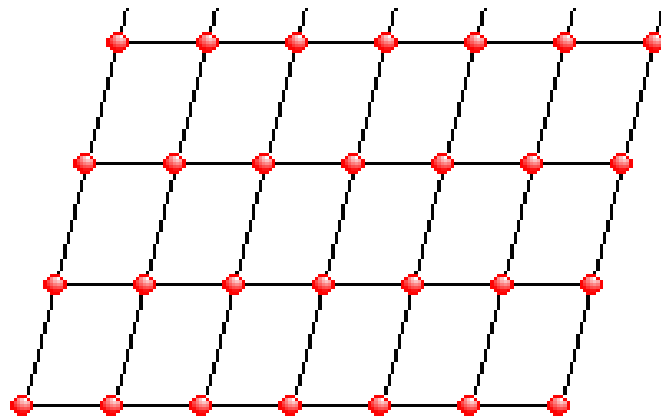
- Don't mix up atoms with lattice points
- Lattice points are infinitesimal points in space
- Lattice points do not necessarily lie at the centre of atoms



Crystal Lattice

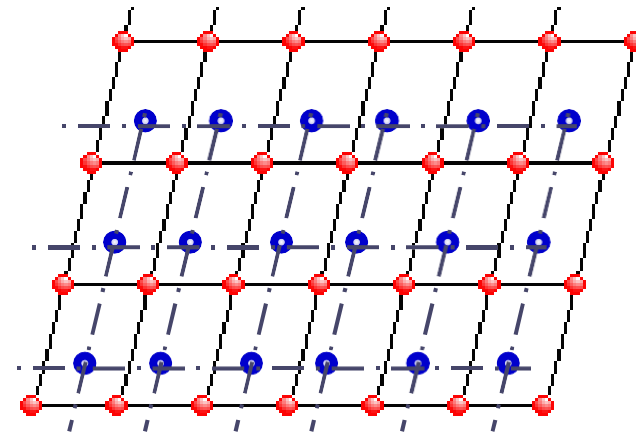
Bravais Lattice (BL)

- All atoms are of the same kind
- All lattice points are equivalent



Non-Bravais Lattice (non-BL)

- Atoms can be of different kind
- Some lattice points are not equivalent
- A combination of two or more BL

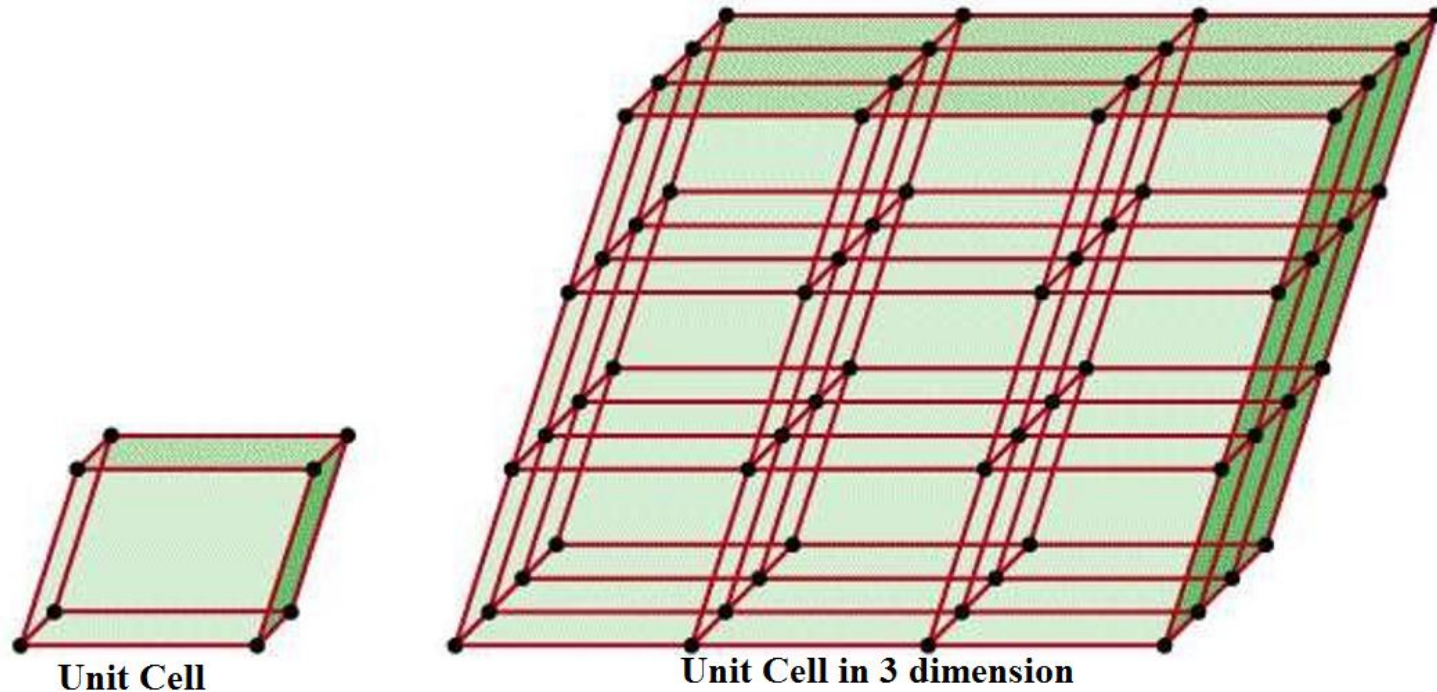


Bravais Lattice

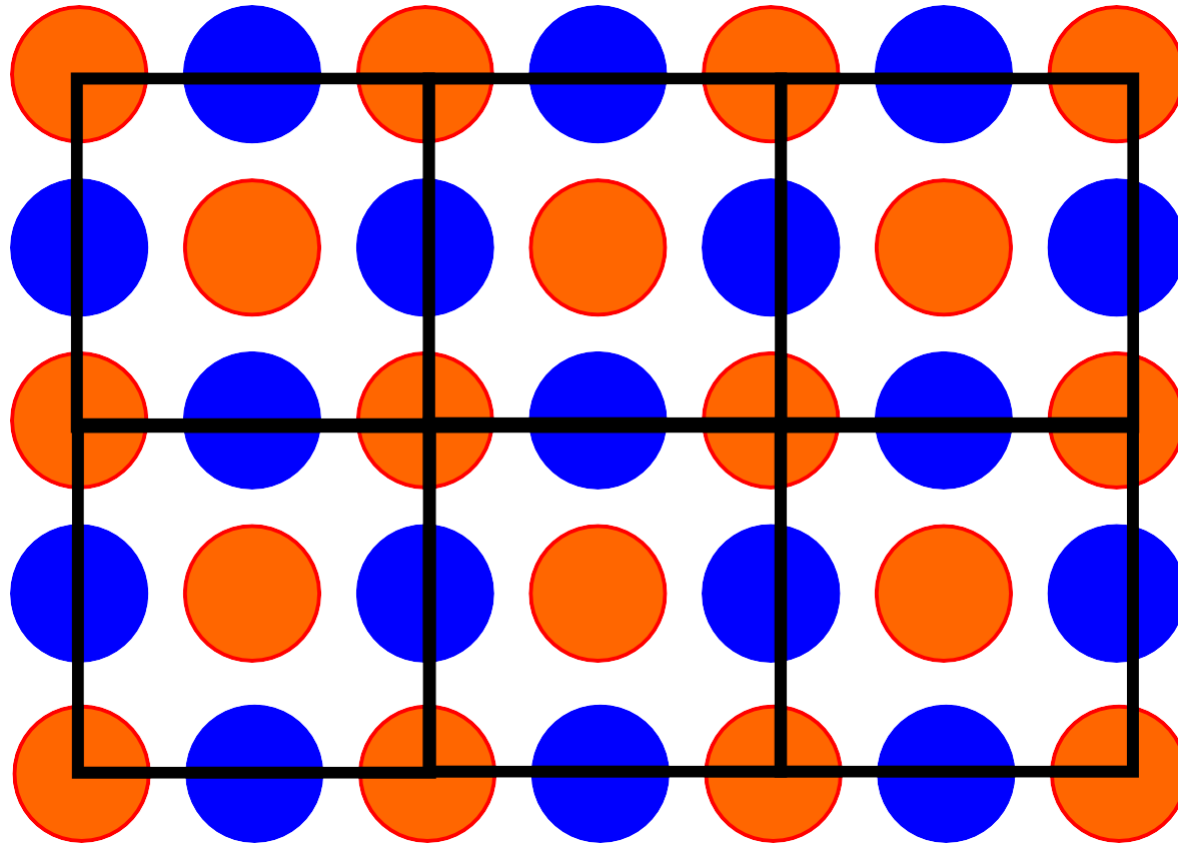
- Bravais lattice, any possible three-dimensional configurations of points used to describe the orderly arrangement of atoms in a crystal.
- Each point represents one or more atoms in the actual crystal, and if the points are connected by lines, a crystal lattice is formed; the lattice is divided into a number of identical blocks, or unit cells, characteristic of the Bravais lattices.
- Lattice points in the Bravais lattice no longer represent a position of a particular atom. Instead a lattice point represents a position in which an atom can be placed. What matters in a Bravais lattice are not the points themselves, but how they are arranged (i.e. symmetry).
- There are 14 possible Bravais lattices.
- The French scientist Auguste Bravais demonstrated in 1850 that only these 14 types of unit cells are compatible with the orderly arrangements of atoms found in crystals.
- The lattices listed by Bravais are divided into six or seven major crystal symmetry systems.
- All crystallographers recognize the isometric, orthorhombic, monoclinic, tetragonal, triclinic, and hexagonal systems ; some, however, define the hexagonal system so as to include the trigonal or rhombohedral system, which is considered a seventh system by others.

Unit Cell

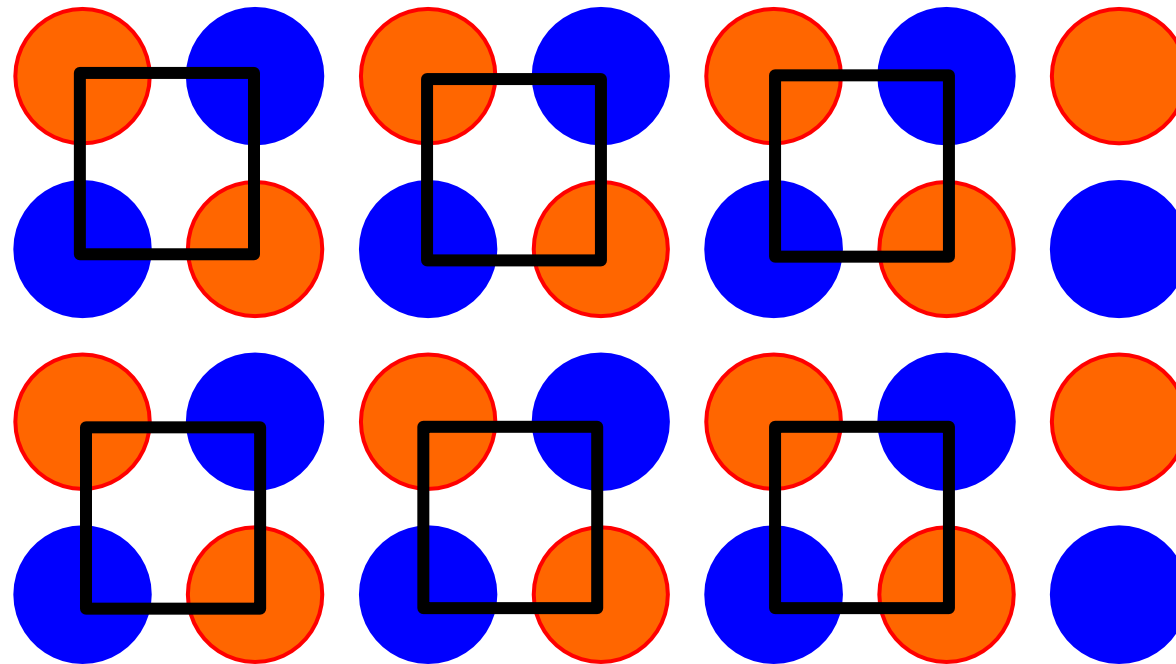
- The unit cell is the simplest repeating unit in the crystal.
- Atoms or group of atoms forming a building block of the smallest acceptable size of the whole volume of a crystal is defined as a unit cell



**This is also a unit cell -
It doesn't matter if you start from Na or Cl**



This is **NOT a unit cell** even though they are all the same
- Empty space is not allowed!



Unit Cell Contents

Counting the number of atoms within the unit cell

Atoms	Shared Between:	Each atom counts:
corner	8 cells	$1/8$
face centre	2 cells	$1/2$
body centre	1 cell	1

lattice type	cell contents
P	1 $[=8 \times 1/8]$
I	2 $[=(8 \times 1/8) + (1 \times 1)]$
F	4 $[=(8 \times 1/8) + (6 \times 1/2)]$

Typical Crystal Structures

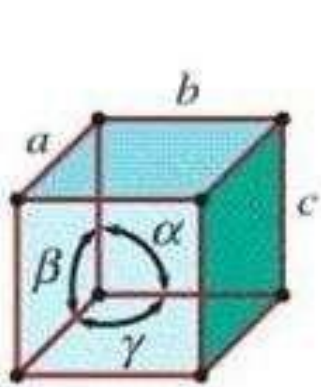
3D – 14 Bravais Lattices And The Seven Crystal System

- There are only seven different shapes of unit cell which can be stacked together to completely fill all space (in 3 dimensions) without overlapping.
- This gives the seven crystal systems, in which all crystal structures can be classified.

The Seven Crystal System

- **Cubic Crystal System (SC, BCC, FCC)**
- **Hexagonal Crystal System (S)**
- **Triclinic Crystal System (S)**
- **Monoclinic Crystal System (S, Base-C)**
- **Orthorhombic Crystal System (S, Base-C, BC, FC)**
- **Tetragonal Crystal System (S, BC)**
- **Trigonal (Rhombohedral) Crystal System (S)**

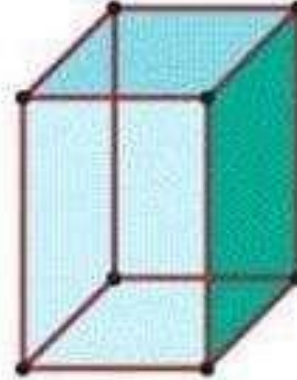
- There are seven unique arrangements, known as crystal systems, which fill in a three dimensional space.



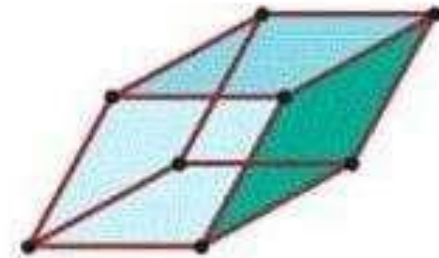
Simple cubic
 $a = b = c$
 $\alpha = \beta = \gamma = 90^\circ$



Tetragonal
 $a = b \neq c$
 $\alpha = \beta = \gamma = 90^\circ$



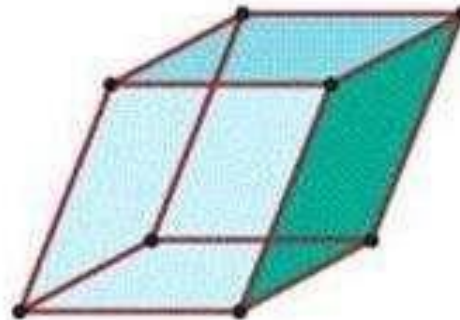
Orthorhombic
 $a \neq b \neq c$
 $\alpha = \beta = \gamma = 90^\circ$



Rhombohedral
 $a = b = c$
 $\alpha = \beta = \gamma \neq 90^\circ$



Monoclinic
 $a \neq b \neq c$
 $\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$



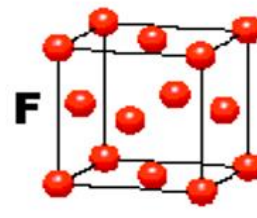
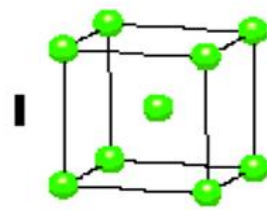
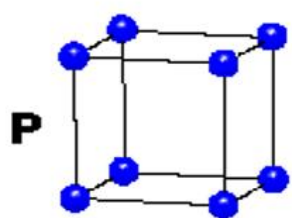
Triclinic
 $a \neq b \neq c$
 $\alpha \neq \beta \neq \gamma \neq 90^\circ$



Hexagonal
 $a = b \neq c$
 $\alpha = \beta = 90^\circ, \gamma = 120^\circ$

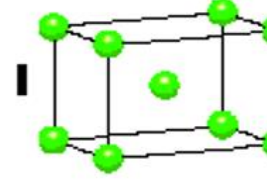
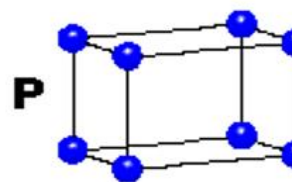
CUBIC

$$a = b = c$$
$$\alpha = \beta = \gamma = 90^\circ$$



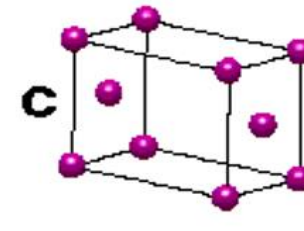
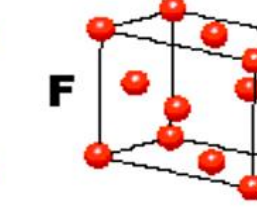
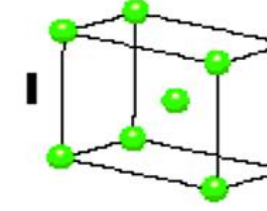
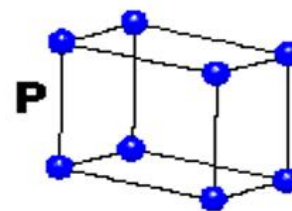
TETRAGONAL

$$a = b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$



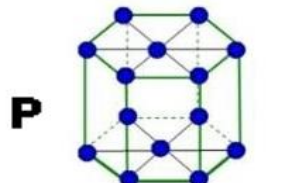
ORTHORHOMBIC

$$a \neq b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$



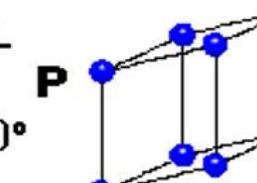
HEXAGONAL

$$a = b \neq c$$
$$\alpha = \beta = 90^\circ$$
$$\gamma = 120^\circ$$



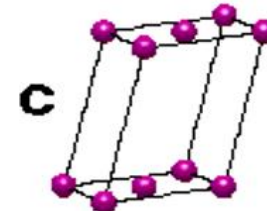
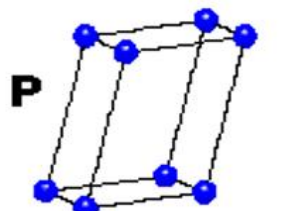
TRIGONAL

$$a = b = c$$
$$\alpha = \beta = \gamma \neq 90^\circ$$



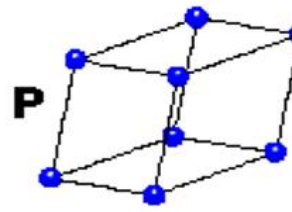
MONOCLINIC

$$a \neq b \neq c$$
$$\alpha = \gamma = 90^\circ$$
$$\beta \neq 120^\circ$$



TRICLINIC

$$a \neq b \neq c$$
$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



4 Types of Unit Cell

P = Primitive

I = Body-Centred

F = Face-Centred

C = Side-Centred

+

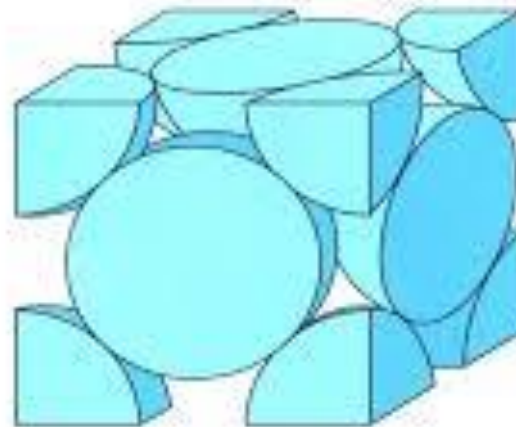
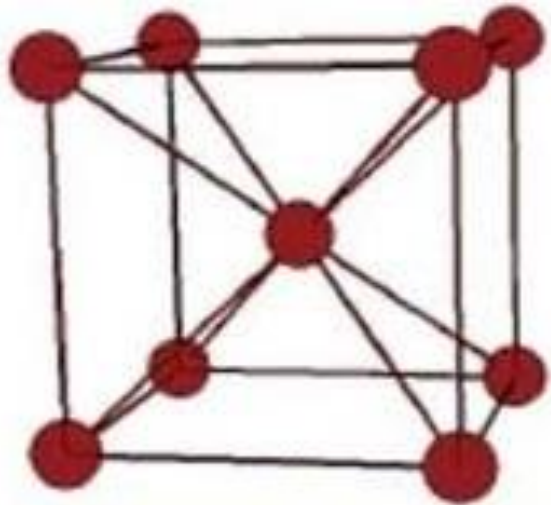
7 Crystal Classes

→ **14 Bravais Lattices**

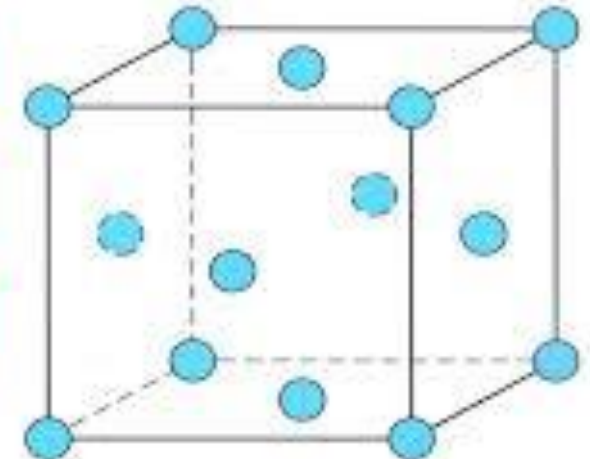
Name	Number of Bravais lattices	© Bart Van Zeghbroeck 2007 Conditions	Primitive	Base-centered	Body-centered	Face-centered
Triclinic	1	$a_1 \neq a_2 \neq a_3, \alpha \neq \beta \neq \gamma$	✓			
Monoclinic	2	$a_1 \neq a_2 \neq a_3, \alpha = \beta = 90^\circ \neq \gamma$	✓	✓		
Orthorhombic	4	$a_1 \neq a_2 \neq a_3, \alpha = \beta = \gamma = 90^\circ$	✓	✓	✓	✓
Tetragonal	2	$a_1 = a_2 \neq a_3$ $\alpha = \beta = \gamma = 90^\circ$	✓		✓	
Cubic	3	$a_1 = a_2 = a_3, \alpha = \beta = \gamma = 90^\circ$	✓		✓	✓
Trigonal	1	$a_1 = a_2 = a_3, \alpha = \beta = \gamma < 120^\circ \neq 90^\circ$	✓			
Hexagonal	1	$a_1 = a_2 \neq a_3, \alpha = \beta = 90^\circ, \gamma = 120^\circ$	✓			

Coordination Number (CN)

- The number of lattice point closest to a given point.
- The Bravais lattice points closest to a given point are the nearest neighbours.
- Because the Bravais lattice is periodic, all points have the same number of nearest neighbours or coordination number. It is a property of the lattice.
 - A simple cubic has coordination number 6;
 - A body-centered cubic lattice, 8;
 - A face-centered cubic lattice, 12.



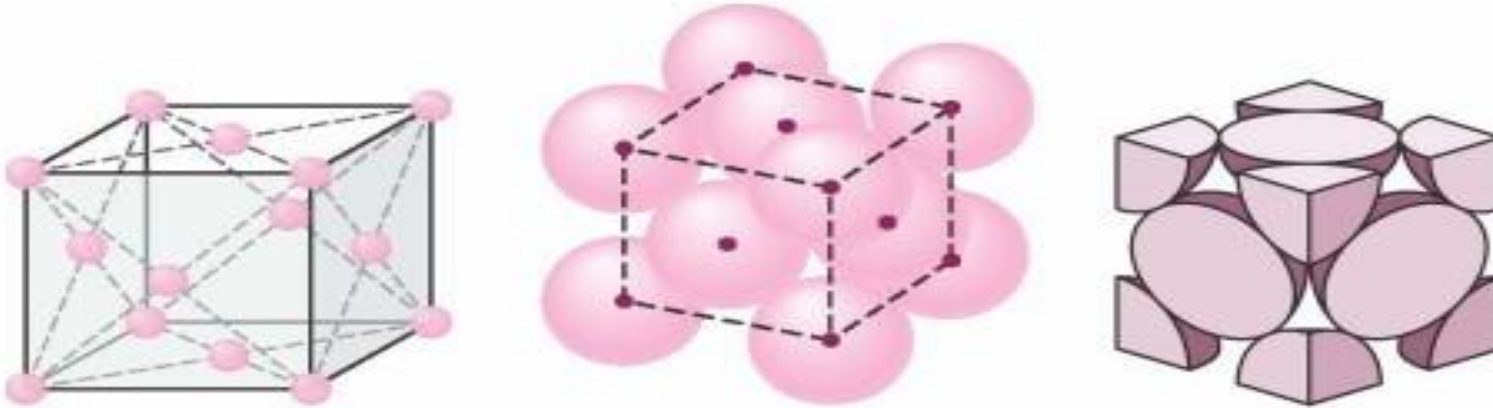
(a)



(b)

Coordination Number (CN)

- ❖ Coordination Number of FCC
- ❖ The corner atom in its own plane touches 4 face centred atoms.
- ❖ In the plane just above, the corner atom has another 4 face centered atoms as its nearest neighbours
- ❖ Similarly, in the plane just below it has 4 more face centered atoms as its nearest neighbours
- ❖ Therefore the no. of nearest neighbours are : $4 + 4 + 4 = 12$



Many of common metals (Cu,Ni,Pb..etc) crystallize in FCC structure.

Atomic Packing Factor

- Atomic Packing Factor (APF) is defined as the volume of atoms within the unit cell divided by the volume of the unitcell.

$$APF = \frac{\text{Volume of Atoms in Unit Cell}}{\text{Volume of Unit Cell}}$$

$$APF = \frac{N_{atom} V_{atom}}{V_{unit\ cell}}$$

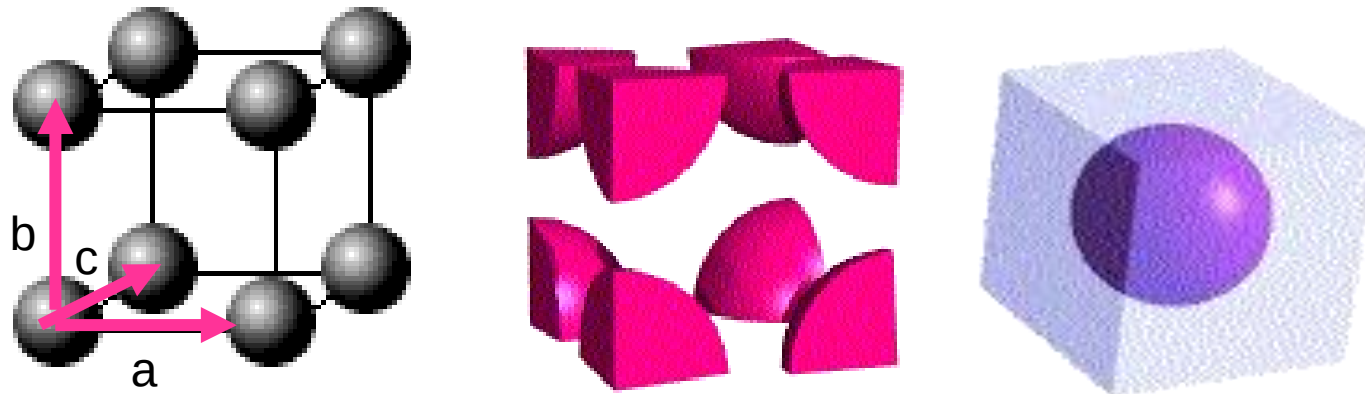
Atomic Packing Factor Of Cubic Crystal System

- $V_{atom} = \frac{4\pi r^3}{3}$; $V_{unit\ cell} = a^3$

- **Therefore, APF = $\frac{N_{atom} \frac{4\pi r^3}{3}}{a^3}$**

Atomic Packing Factor Of Simple Cubic (SC)

- Simple Cubic has one lattice point, so its primitive cell.
- In the unit cell on the left, the atoms at the corners are cut because only a portion (in this case $1/8$) belongs to that cell. The rest of the atom belongs to neighboring cells.
- Coordination number of simple cubic is 6.

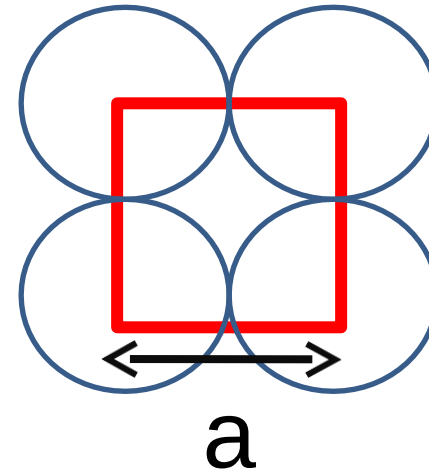


Atomic Packing Factor Of Simple Cubic (SC)

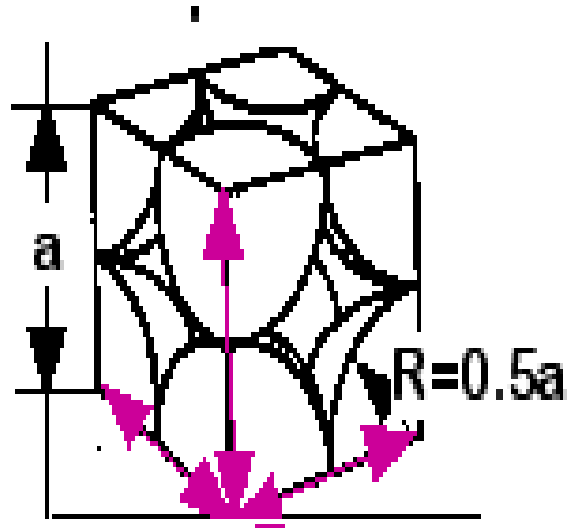
- Atomic radius for simple cubic is half the distance between any two nearest neighbors in the given crystal structure.
- It is expressed in terms of cube edge 'a'

$$a = 2r, \quad r = a/2$$

$$\text{Atomic Radius, } r = 0.5a$$



Atomic Packing Factor Of Simple Cubic (SC)



contains $8 \times 1/8 =$
1 atom/unit cell

APF = 0.52 for simple cubic

$$\text{APF} = \frac{\text{atom/unit cell} \times \frac{4}{3} \pi (0.5a)^3}{a^3}$$

The diagram shows the calculation of the Atomic Packing Factor (APF) for a simple cubic unit cell. The formula is presented as a fraction. The numerator is enclosed in a green box, and the denominator is enclosed in a blue box. Arrows point from the labels to the corresponding parts of the formula:

- A green arrow points from "atom/unit cell" to the "1" in the numerator.
- A brown arrow points from "volume/atom" to the $\frac{4}{3} \pi (0.5a)^3$ term in the numerator.
- A blue arrow points from "volume/unit cell" to the a^3 term in the denominator.

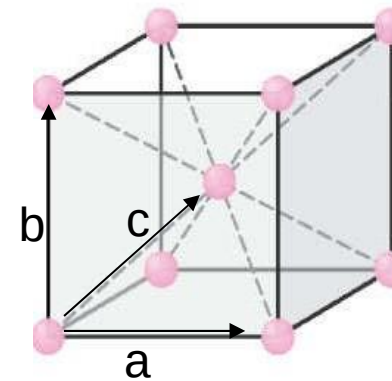
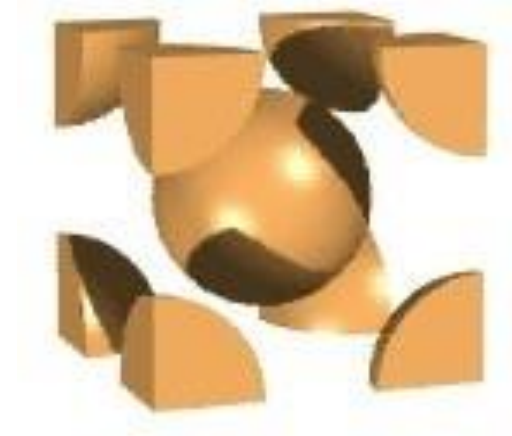
Atomic Packing Factor Of Simple Cubic (SC)

- **APF = 0.52**
- **That means that the percentage of packing is 52%**
- **Thus, 52% of the volume of the simple cubic unit cell is occupied by atoms and the remaining 48% volume of unit cell is vacant or void space.**

Atomic Packing Factor Of Body Centered Cubic (BCC)

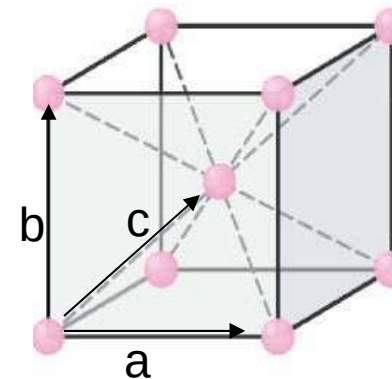
- BCC structure has 8 corner atoms and 1 body centreatom.
- Each corner atom is shared by 8 unitcells.
- The center atom is not shared by any of the unitcells.
- Therefore, **Number of atoms per unitcell**

$$N = \frac{1}{8} \times 8 + 1 = 2$$



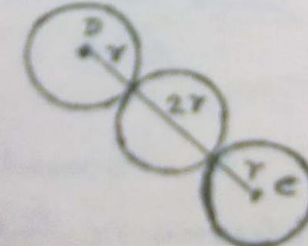
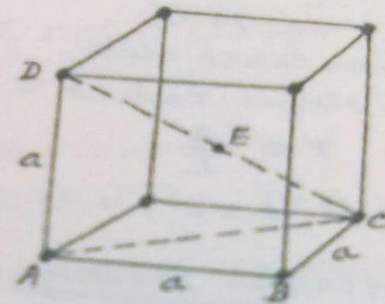
Atomic Packing Factor Of Body Centered Cubic (BCC)

- BCC has two lattice points so BCC is a non-primitive cell.
- BCC has eight nearest neighbors. Each atom is in contact with its neighbors only along the body-diagonal directions.
- **Hence, the coordination no. for BCC unit cell is 8**
- Many metals (Fe, Li, Na..etc), including the alkalis and several transition elements choose the BCC structure.



Atomic Radius For BCC Unit Cell

From figure,
 $AB = BC = AD = a$



$CD = 4r$

From triangle ABC

$$AC^2 = AB^2 + BC^2 = a^2 + a^2 = 2a^2$$
$$AC = a\sqrt{2}$$

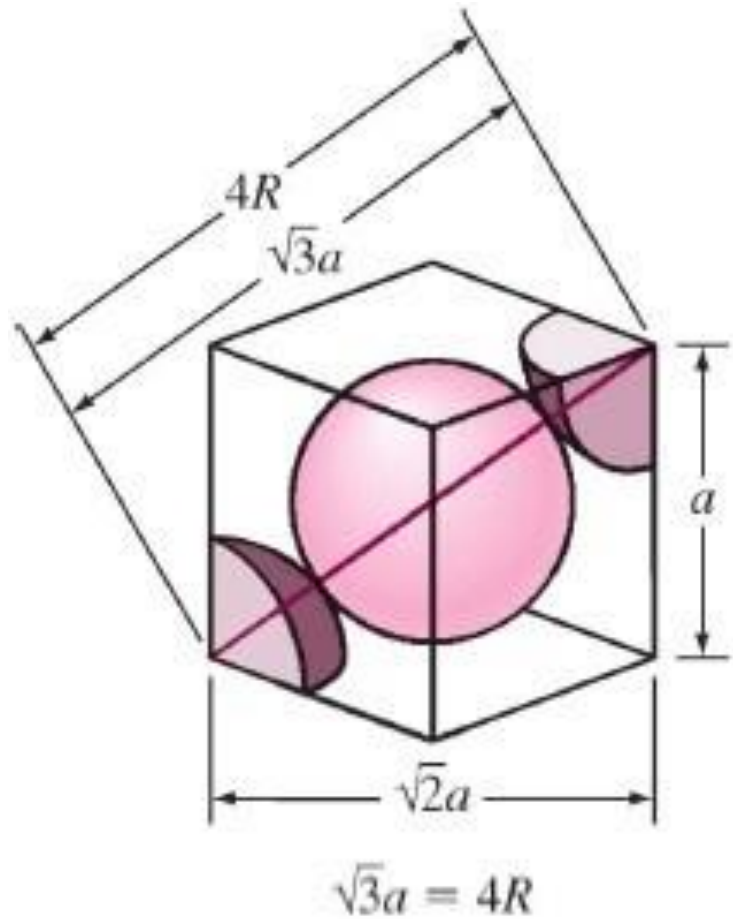
From triangle ACD

$$CD^2 = AC^2 + AD^2 = 2a^2 + a^2 = 3a^2$$
$$(4r)^2 = 3a^2$$
$$16r^2 = 3a^2$$
$$r^2 = 3a^2 / 16$$

atomic radius $(r) = a\sqrt{3} / 4$

$$r = a \frac{\sqrt{3}}{4}$$

Atomic Packing Factor Of Body Centered Cubic (BCC)

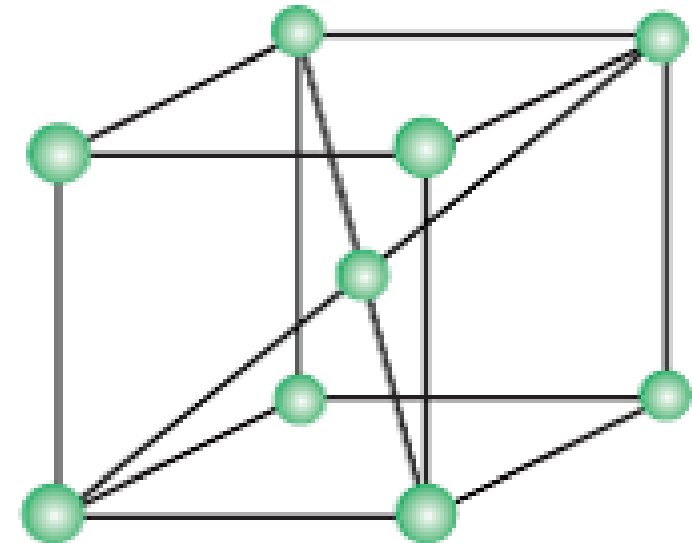


$$APF_{BCC} = \frac{N_{atom} V_{atom}}{V_{unit\ cell}} = 0.68$$

$$APF = \frac{\overbrace{2}^{\substack{\text{atom} \\ \text{unit cell}}} \cdot \overbrace{\frac{4}{3} \pi (0.433a)^3}^{\substack{\text{volume} \\ \text{atom}}}}{\underbrace{a^3}_{\substack{\text{volume} \\ \text{unit cell}}}}$$

Atomic Packing Factor Of Body Centered Cubic (BCC)

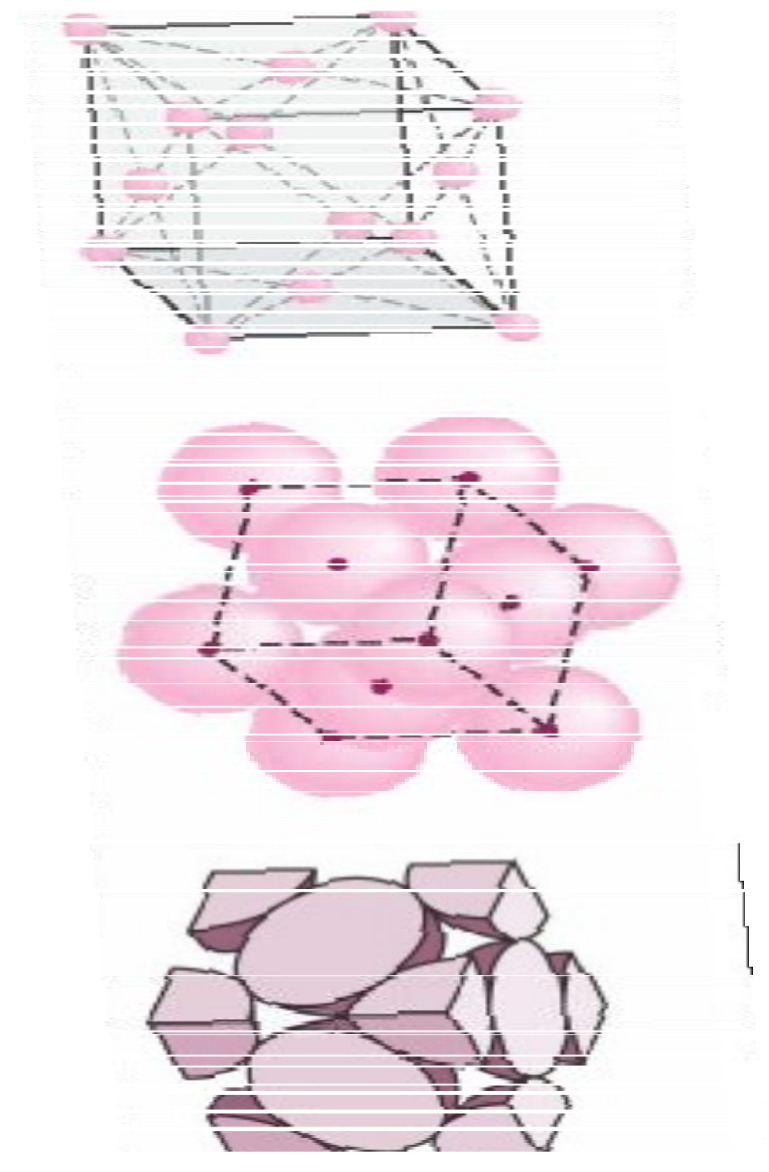
- The percentage of packing for BCC structure is 68%
- Thus , 68% of the volume of body centered cubic cell is occupied by atoms and the remaining 32% of the volume is vacant or void space



Atomic Packing Factor Of Face Centered Cubic (FCC)

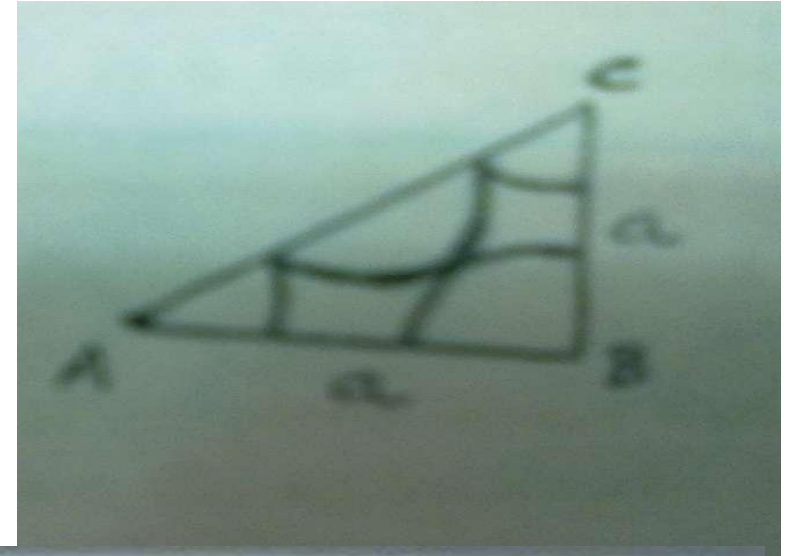
- FCC structure has 8 corner atoms and 6 face centre atoms.
- Each corner atom is shared by 8 unit cells.
- Each face centered atom is shared by 2 unit cells.
- Therefore, **Number of atoms present in unit cell is**

$$\begin{aligned} N &= (1/8 \times 8) + (1/2 \times 6) \\ &= 1 + 3 \\ &= 4 \end{aligned}$$



Atomic Packing Factor Of Face Centered Cubic (FCC)

Atomic radius, $r = a \frac{\sqrt{2}}{4}$



$$AB = BC = a \quad \text{and} \quad AC = 4r$$

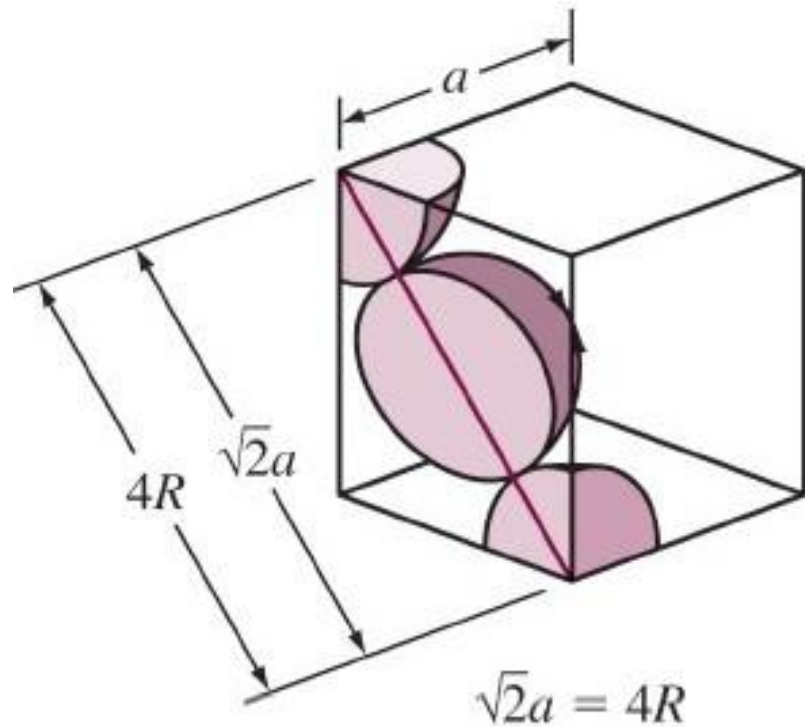
$$\text{From triangle ABC, } AC^2 = AB^2 + BC^2 = a^2 + a^2 = 2a^2$$

$$AC = a\sqrt{2}$$

$$(4r) = a\sqrt{2}$$

$$\text{atomic radius } (r) = a\sqrt{2}/4$$

Atomic Packing Factor Of Face Centered Cubic (FCC)



$$APF_{FCC} = \frac{N_{atom} V_{atom}}{V_{unit\ cell}} = 0.74$$

$$APF = \frac{\overbrace{4}^{\text{atom}} \overbrace{\frac{4}{3} \pi (0.354a)^3}^{\text{volume atom}}}{\underbrace{a^3}_{\text{volume unit cell}}}$$

Atomic Packing Factor Of Face Centered Cubic (FCC)

- The percentage of packing for BCC structure is 74%
- Thus , 74% of the volume of body centered cubic cell is occupied by atoms and the remaining 26% of the volume is vacant or void space

Miller indices

- Certain directions and planes within a unit cell carry particular significance.
- For example, crystallographic directions are used to indicate a particular orientation of a single crystal. Also metals deform along the planes atoms which are most tightly packed.
- A set of three integers or indices known as Miller indices are the shorthand notation used to describe crystallographic directions and planes.
- The basis for determining the index values is the unit cell with a co-ordinate system having three axes along the edges situated at one of the corners.
- For some unit cells, (rhombohedral, monoclinic, triclinic and hexagonal) the three axes are not mutually perpendicular as in the familiar cartesian system.
- A crystallographic direction is defined as a line or vector between two points.

Miller indices

- ❖ **Miller indices, group of three numbers that indicates the orientation of a plane or set of parallel planes of atoms in a crystal.**
- ❖ **If each atom in the crystal is represented by a point and these points are connected by lines, the resulting lattice may be divided into a number of identical blocks, or unit cells; the intersecting edges of one of the unit cells defines a set of crystallographic axes, and the Miller indices are determined by the intersection of the plane with these axes.**
- ❖ **The reciprocals of these intercepts are computed, and fractions are cleared to give the three Miller indices (hkl).**

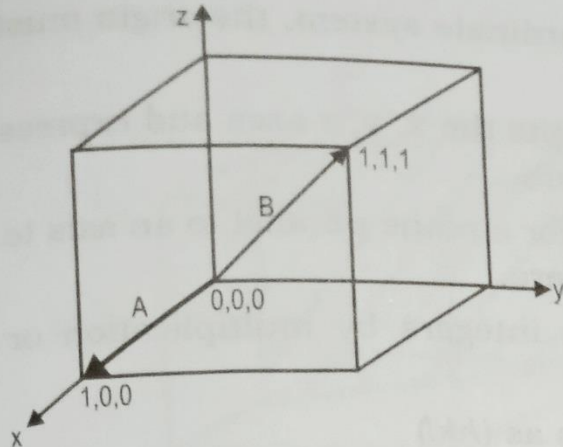
Miller indices

❖ Following are the steps for determination of the Miller indices of a direction.

- 1) Determine the co-ordinates of two points that lie on the direction, using a right handed co-ordinate system.
- 2) Subtract the co-ordinates of tail point from those of the head point and express the same in terms of the unit cell dimensions.
- 3) Multiply or divide these numbers by a common factor to reduce them to the smallest integers.
- 4) The three indices are enclosed in square brackets as $[uvw]$. A negative integer is represented with a bar over the number.

Miller indices

Q: Determine the Miller indices of directions A, B, C and D as shown in the cubic unit cells.



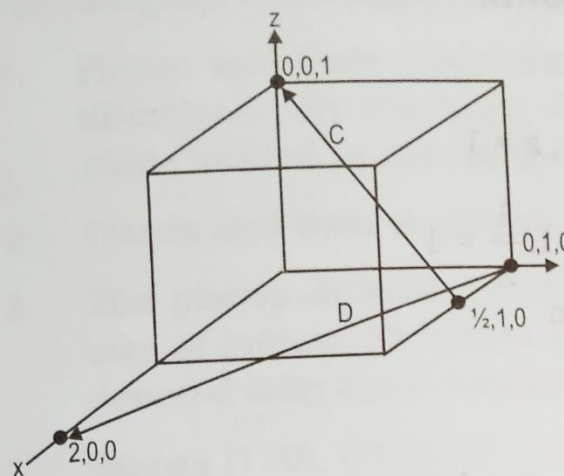
A 3D coordinate system with x, y, and z axes. A cube is drawn with origin at (0,0,0). Direction A is a vector from (0,0,0) to (1,0,0) on the x-axis. Direction B is a vector from (0,0,0) to (1,1,1) at the opposite corner of the cube.

Direction A

- Points are 1, 0, 0 and 0, 0, 0
- $1, 0, 0 - 0, 0, 0 = 1, 0, 0$
- No fractions to clear
- $[1\ 0\ 0]$

Direction B

- Points are 1, 1, 1 and 0, 0, 0
- $1, 1, 1 - 0, 0, 0 = 1, 1, 1$
- No fractions to clear
- $[1\ 1\ 1]$



A 3D coordinate system with x, y, and z axes. A cube is drawn with origin at (0,0,0). Direction C is a vector from (0,0,1) on the z-axis to (1/2, 1, 0) on the edge between x and y axes. Direction D is a vector from (2,0,0) on the x-axis to (0,1,0) on the y-axis.

Direction C

- Points are 0, 0, 1 and $\frac{1}{2}, 1, 0$
- $0, 0, 1 - \frac{1}{2}, 1, 0 = -\frac{1}{2}, -1, 1$
- $2(-\frac{1}{2}, -1, 1) = -1, -2, 2$
- $[\bar{1}\ \bar{2}\ 2]$

Direction D

- Points are 0, 1, 0 and 2, 0, 0
- $2, 0, 0 - 0, 1, 0 = 2, -1, 0$
- No fractions
- $[2\ \bar{1}\ 0]$

Miller indices

- ❖ Following points are to be noted about the use of Miller indices for directions.
 - As the directions are vectors, a direction and its negative are not identical. The directions $[100]$ and $[\bar{1}00]$ represent the same line, but opposite directions.
 - A direction and its multiple are identical; $[120]$ and $[240]$ represent the same direction.
 - In any crystal system, there are certain groups of directions which are equivalent and are called a family of directions.
 - In cubic system, corresponding to $[100]$ direction a family of equivalent directions are obtained by interchanging the positions and signs of indices.
 - The index when enclosed in angle brackets represents a family.

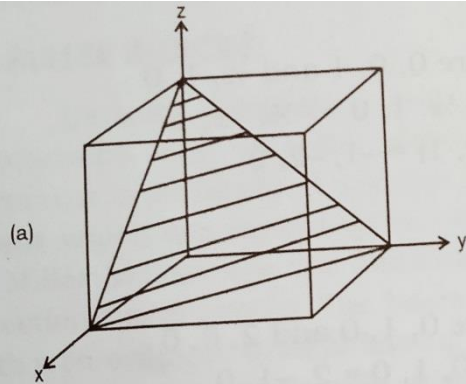
$$\langle 100 \rangle = [100], [\bar{1}00], [010], [0\bar{1}0], [001] \text{ and } [00\bar{1}]$$

Miller indices for Planes

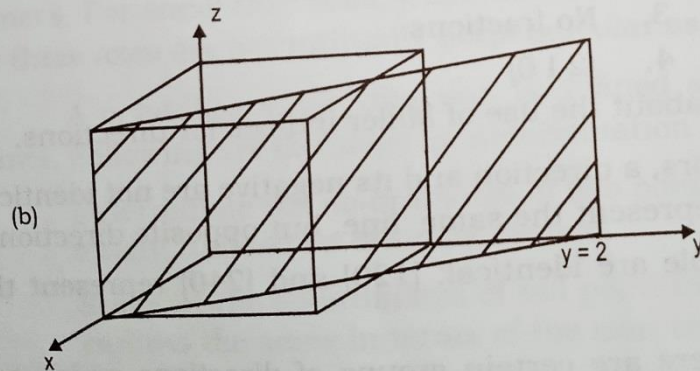
❖ Orientations of planes for a crystal structure are represented in a similar manner. Here also the basis is the unit cell, with a three axis co-ordinate system oriented along the edges of the unit cell. The Miller index for planes is specified as (hkl) which is determined using the following procedure.

- 1) If the plane passes through the origin of co-ordinate system, the origin must be moved to another corner of unit cell.
- 2) Identify the points at which the plane intercepts the x , y , z axis and express the intercepts in terms of the lattice parameters.
- 3) Take reciprocals of these numbers. Intercept for a plane parallel to an axis to be taken as infinity with reciprocal equal to zero.
- 4) Change these numbers to a set of smallest integers by multiplication or division by a common factor.
- 5) Enclose the integer indices within parentheses as (hkl)

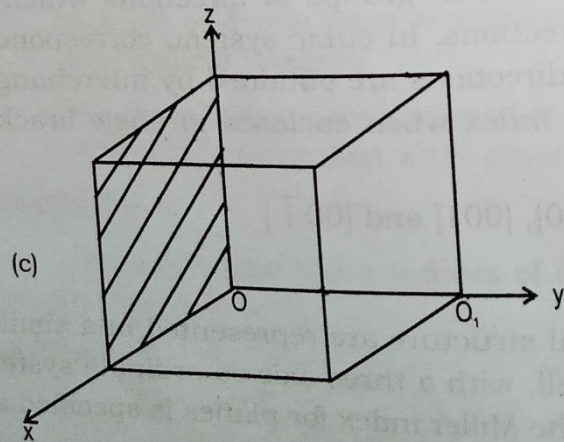
Q: Determine the Miller indices of the planes shown.



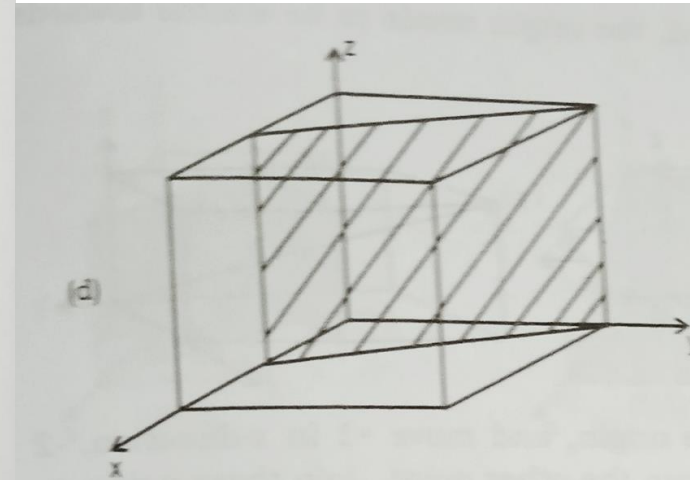
1. $x = 1, y = 1, z = 1$
2. $\frac{1}{x} = 1, \frac{1}{y} = 1, \frac{1}{z} = 1$
3. No fraction
4. $(1 \ 1 \ 1)$



1. $x = 1, y = 2, z = \infty$
plane is parallel to z axis
2. $\frac{1}{x} = 1, \frac{1}{y} = \frac{1}{2}, \frac{1}{z} = 0$
3. Multiply by 2
4. $(2 \ 1 \ 0)$



- Origin to be moved from O to O₁
1. $x = \infty, y = -1, z = \infty$
 2. $\frac{1}{x} = 0, \frac{1}{y} = -1, \frac{1}{z} = 0$
 3. No fractions to clear
 4. $(0 \ \bar{1} \ 0)$



1. $x = \frac{1}{2}, y = 1, z = \infty$
2. $\frac{1}{x} = 2, \frac{1}{y} = 1, \frac{1}{z} = 0$
3. No fraction
4. $(2 \ 1 \ 0)$

Miller indices for Planes

❖ Following are some important aspects of the Miller indices for planes

- 1) Planes and their negatives are identical: $(010) = \bar{0}10$ unlike the case of directions. By changing the signs of all the indices of a plane, we obtain a plane located at the same distance on the other side of the origin.
- 2) Planes and their multiples are not identical (unlike the case of directions)
- 3) The planes defined by Miller indices obtained by changing the order and sign of indices represent a family of planes. This group of similar planes are denoted with curly brackets. The family $\{110\}$ in cubic system consists of the planes $(110), (101), (011), (\bar{1}10), (1\bar{1}0), (10\bar{1}), (\bar{1}01), (01\bar{1}), (0\bar{1}1), (\bar{1}10), (0\bar{1}\bar{1})$ and $(\bar{1}0\bar{1})$.
- 4) In cubic system, a plane and a direction having the same indices are perpendicular to each other. Hence the direction $[110]$ is perpendicular to the plane (110) , in a cubic unit cell.
- 5) Parallel planes have the same Miller indices. Distance between two adjacent parallel planes with same Miller index (hkl) is called interplanar spacing. Interplanar spacing in cubic crystals is given by

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}, \text{ where } a \text{ is the lattice parameter.}$$

Miller indices- Notation Summary

- ✓ (h,k,l) represents a point – note the exclusive use of commas.
- ✓ Negative numbers/directions are denoted with a bar on top of the number
- ✓ $[hkl]$ represents a direction
- ✓ $\langle hkl \rangle$ represents a family of directions
- ✓ (hkl) represents a plane
- ✓ $\{hkl\}$ represents a family of planes

Importance of Miller Indices

❖ **In Materials Science it is important to have a notation system for atomic planes since these planes influence**

✓ **Optical properties**

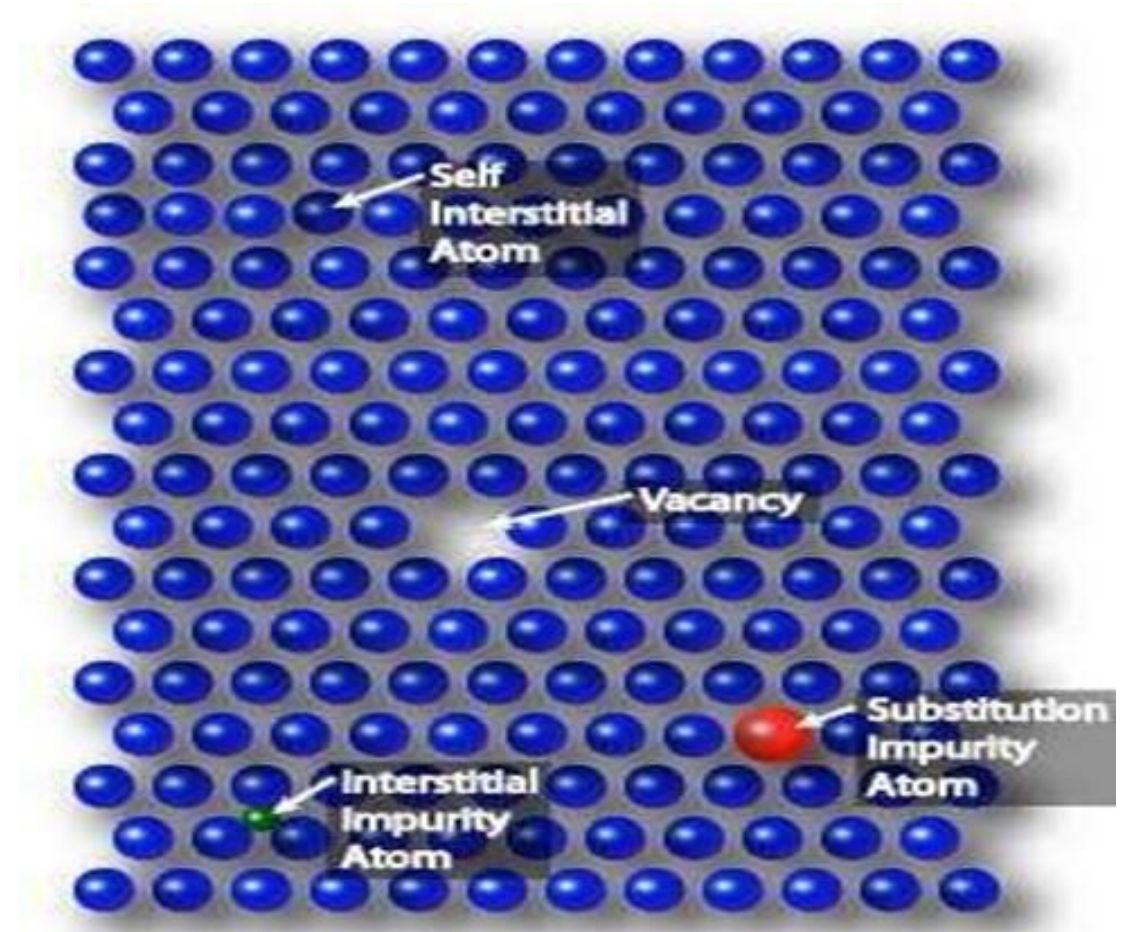
✓ **Reactivity**

✓ **Surface tension**

✓ **Dislocations**

Defects in Solids

- Crystalline material has a crystal structure in which the atoms are positioned in perfect ordered pattern which is repetitive over large atomic distance.
- Defects have a profound impact on the macroscopic properties of materials.
- Sometimes defects are deliberately created to improve properties of crystal.
- Many of the important properties of materials are due to the presence of these imperfections.
- Imperfections in the regular geometrical arrangement of the atoms in a crystalline solid.
- The defects are basically irregularities in the arrangement of constituent particles.



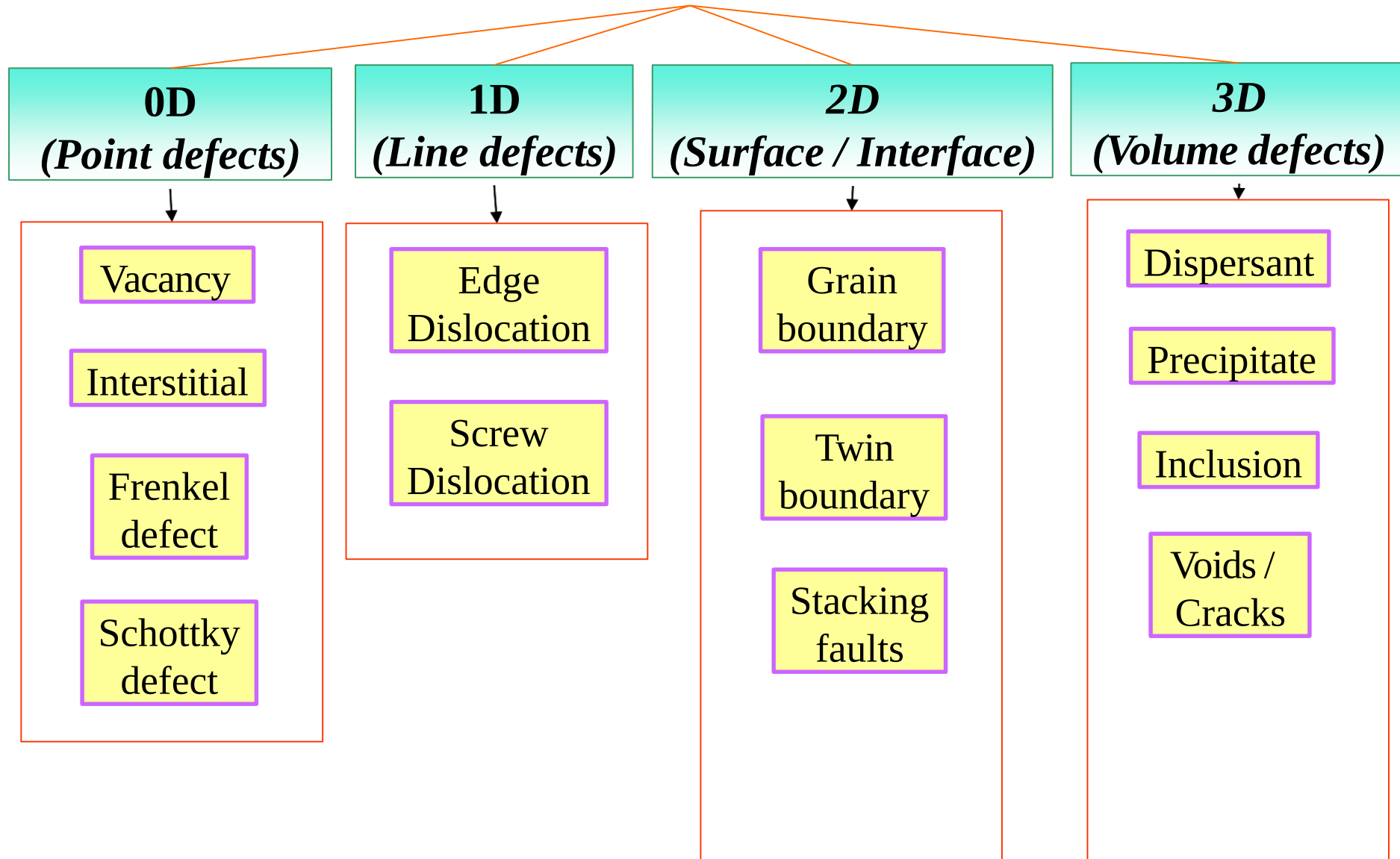
Crystalline Imperfections

“defects” lower the energy of a crystal & make it more stable

✓ always have vacancies and impurities, to some extent

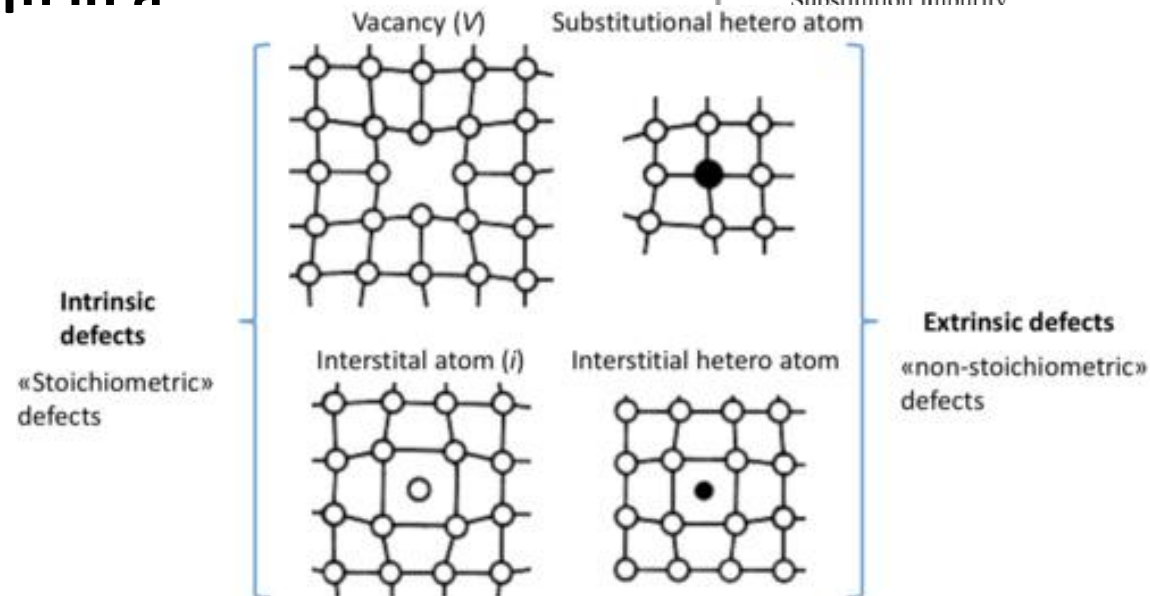
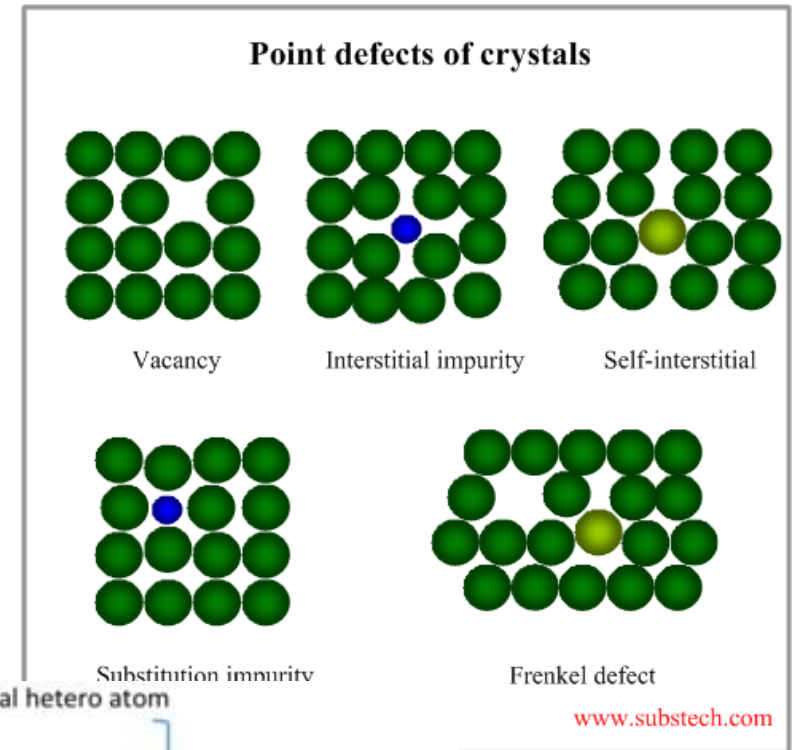
- Defect does not necessarily imply a bad thing
 - addition of C to Fe to make steel
 - addition of Cu to Ni to make thermocouple wires
 - addition of Ge to Si to make thermoelectric materials
 - addition of Cr to Fe for corrosion resistance
 - introduction of grain boundaries to strengthen materials
 - and so on
- *“Defect” (in this context) can be either desirable or undesirable. In general, a defect simply refers to a disruption in the crystalline order of an otherwise periodic material.*

CLASSIFICATION OF DEFECTS BASED ON DIMENSIONALITY



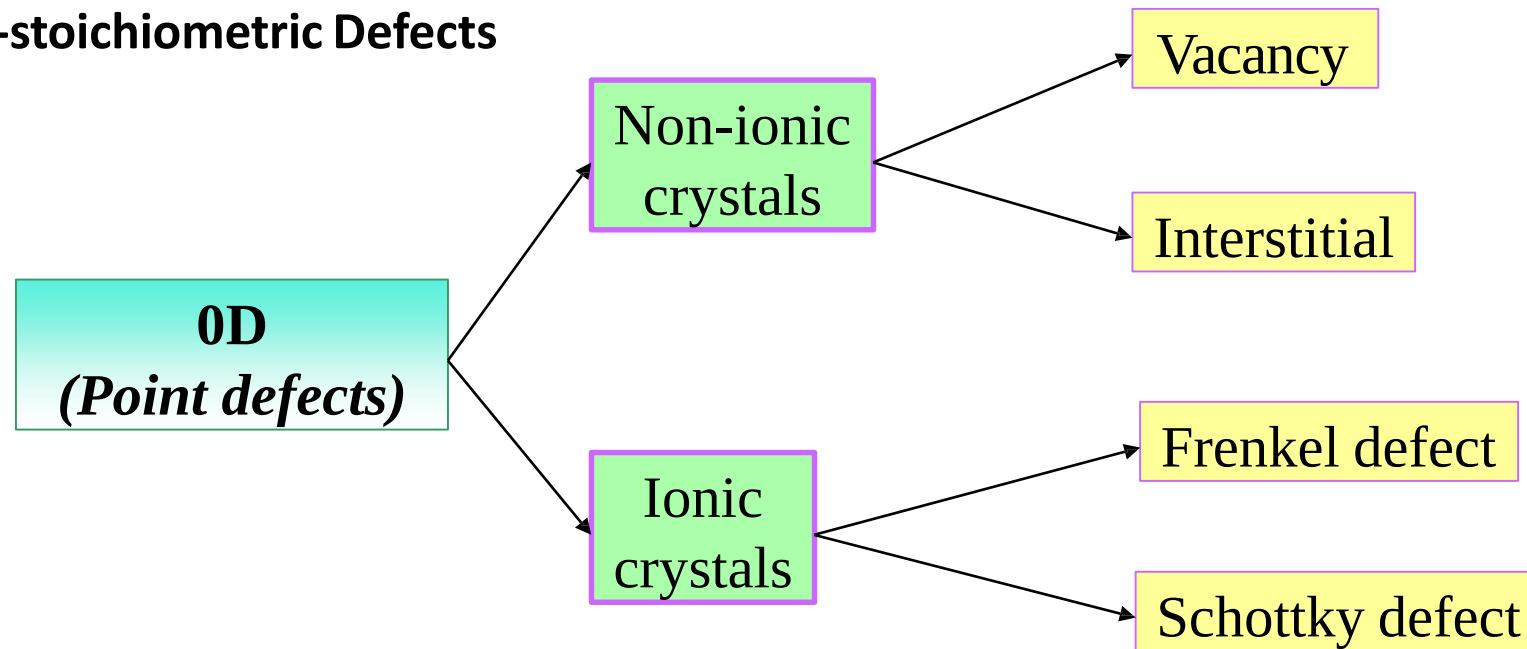
Point Defects

- Atoms in solid possess vibrational energy, some atoms have sufficient energy to break the bonds which hold them in equilibrium position. Hence once the atoms are free they give rise to Point Defects.
- Point Defects are the irregularities or deviations from ideal arrangement around a point or an atom in a crystalline substance.
- ❖ **Classes of point defects:**
 - **Intrinsic defects.**
 - ✓ **Vacancy**
 - ✓ **Interstitial**
 - **Extrinsic defects**
 - ✓ **Substitution**
 - ✓ **Interstitial**



Point Defects

- ❖ Point Defects are the irregularities or deviations from ideal arrangement around a point or an atom in a crystalline substance.
- ❖ Point defects can be classified into 3 types ;
 - (a)Stoichiometric Defect
 - (b)Impurity Defects
 - (c)Non-stoichiometric Defects



Vacancy Defect

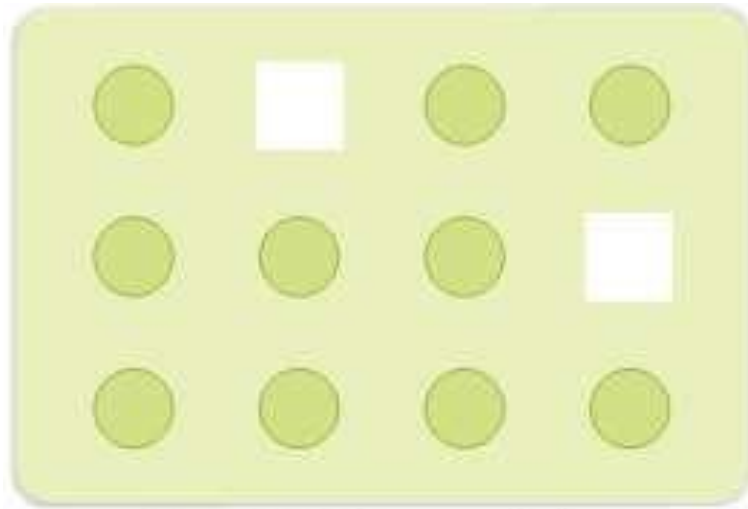
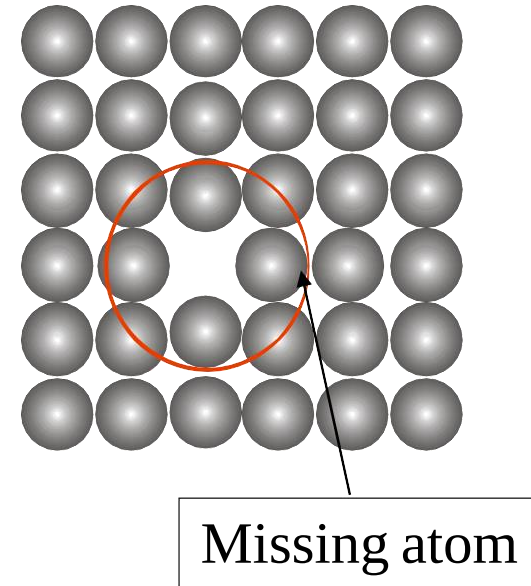
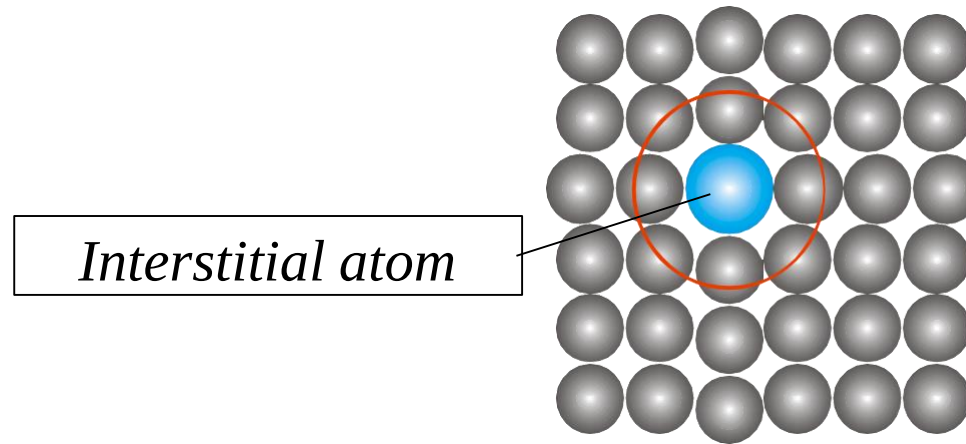


Fig. 1.23: Vacancy defects



- ☐ Atom missing from an atomic site
- ☐ Occur due to imperfect packing during crystallisation
- ☐ This results in decrease in density of the substance
- ☐ Number of vacancy defects depend on temperature

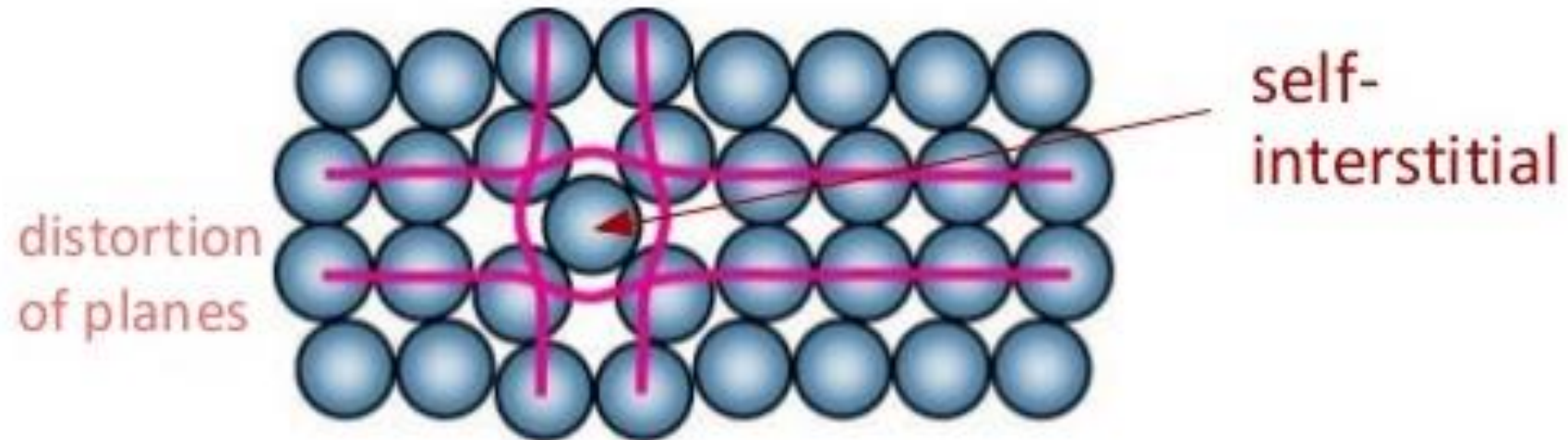
Interstitial Defect



- ☐ When some constituent particles (atoms or molecules) occupy an interstitial site, the crystal is said to have interstitial defect.
- ☐ Addition of an extra atom within a crystal structure
- ☐ This defect increases the density of the substance
- ☐ Causes atomic distortion
- ☐ Vacancy and interstitials are inverse phenomena

Self-interstitial

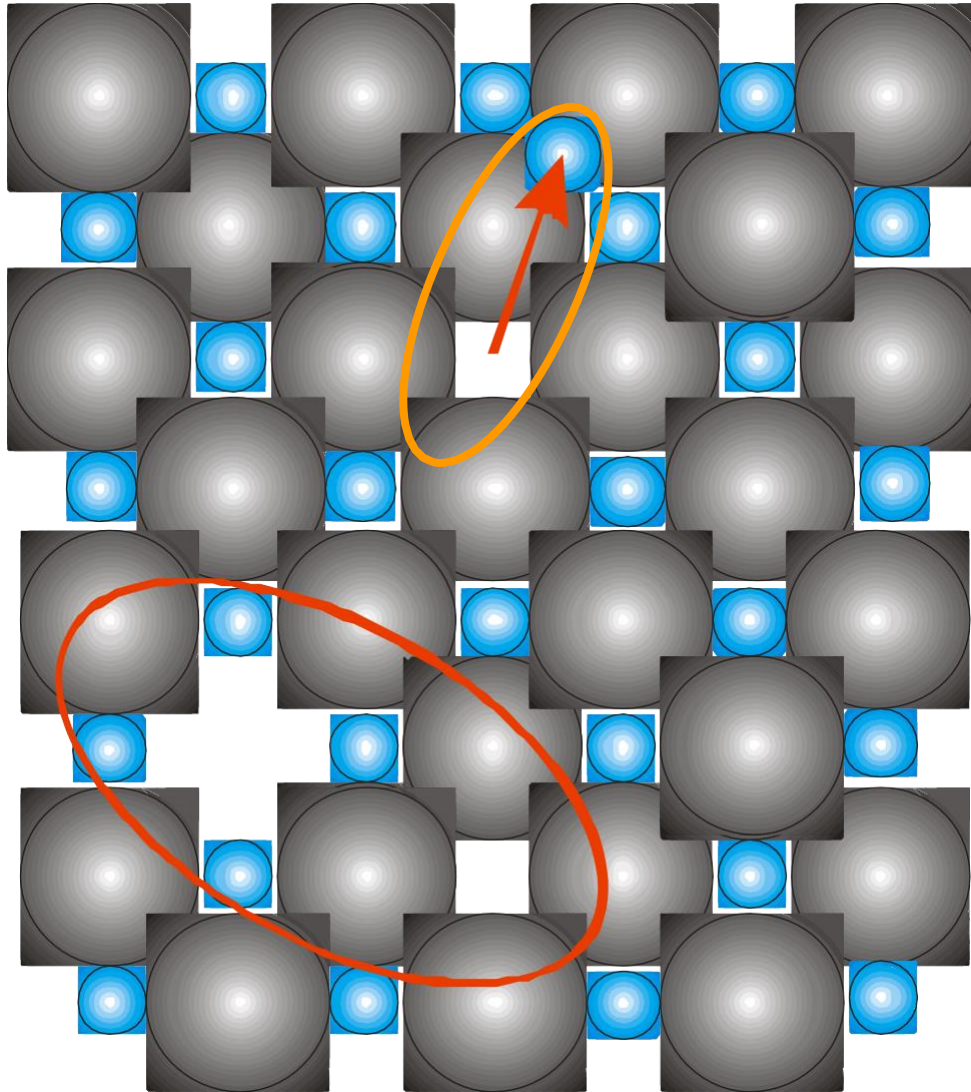
- If the matrix atom occupies its own interstitial site, the defect is called Self Interstitial.
- Extra atoms positioned between atomic sites.
- Self-interstitials in metals introduce large distortions in the surrounding lattice.



Point Defects In Ionic Solids

- ❖ **Ionic solid must always maintain electrical neutrality , rather than simple vacancy or interstitial defects, and these are Frenkel defect and Schottky defect**
- **Schottky Defects**
- ✓ **When cation vacancy is associated with anion vacancy, the defect is called Schottky Defect.**
- **Frenkel Defects**
- ✓ **When an ion leaves its regular site and occupy nearby interstitial site it gives rise to two defects i.e. one vacancy and other self interstitial these two defects are called as Frenkel Defects.**

Defects In Ionic Solids



Frenkel
defect

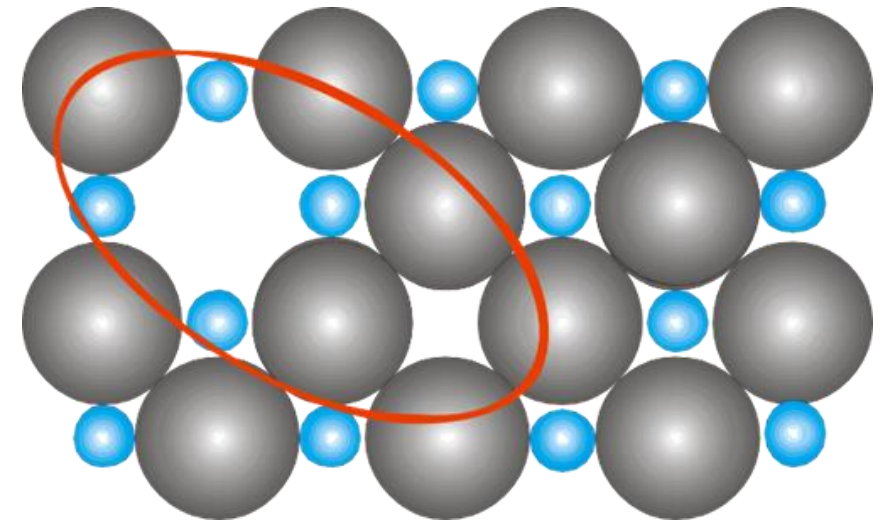
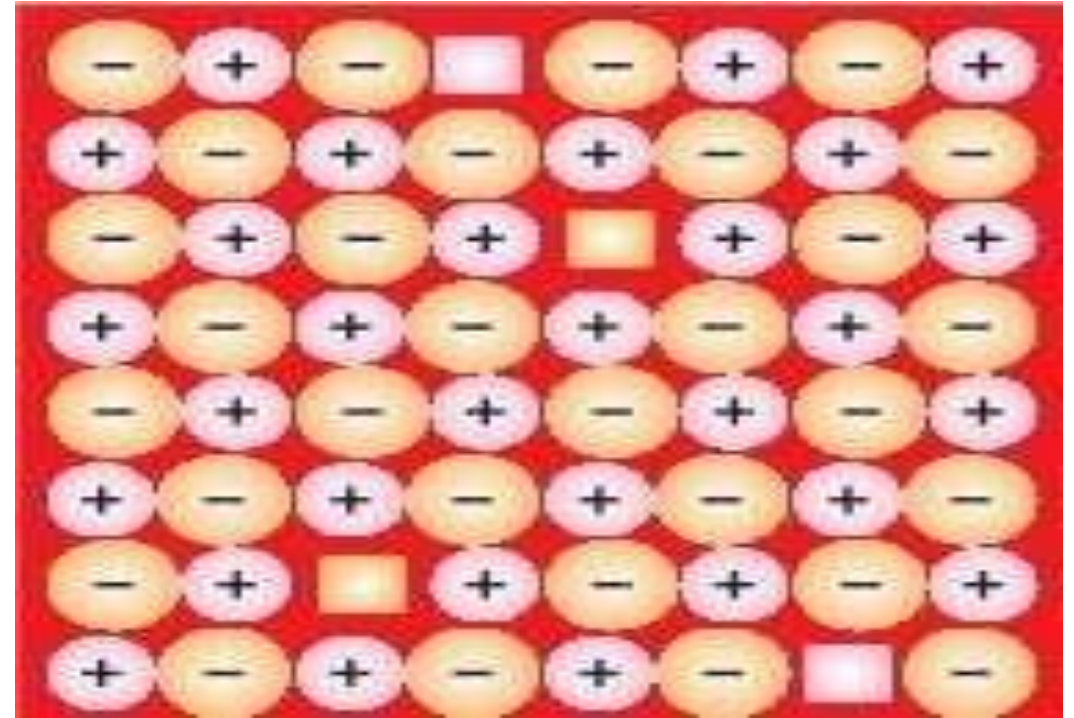
Cation vacancy
+
cation interstitial

Schottky
defect

Cation vacancy
+
anion vacancy

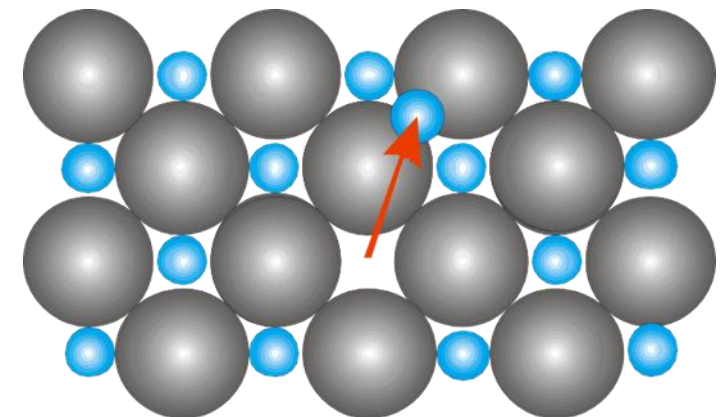
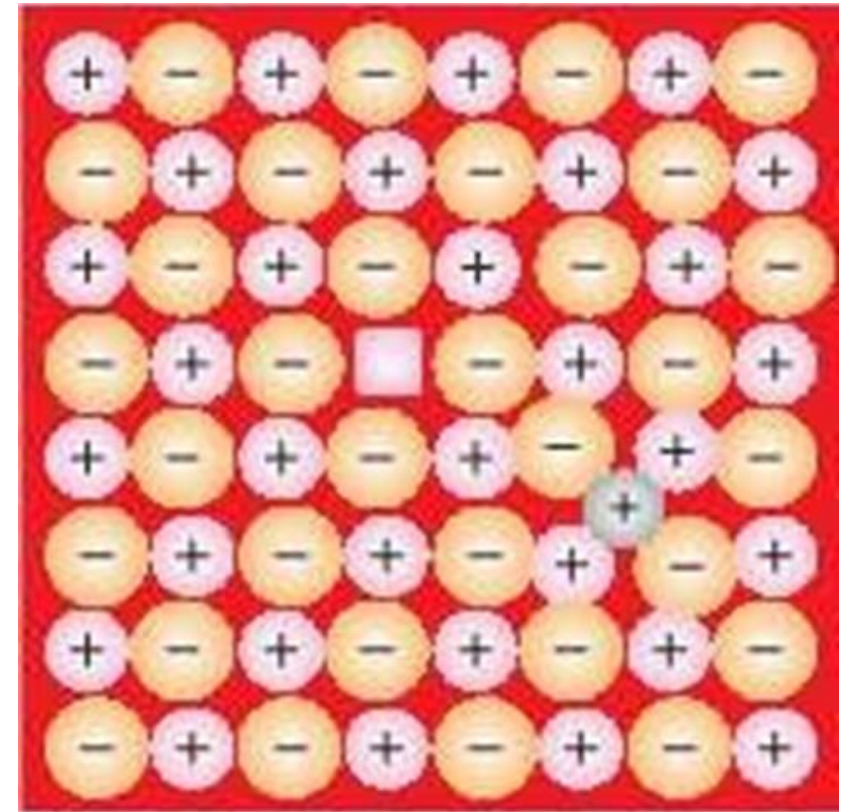
Schottky Defect

- ❖ It is basically a vacancy defect in ionic solids.
- ❖ In order to maintain electrical neutrality, the number of missing cations and anions are equal
- ❖ Like simple vacancy defect, Schottky defect also decreases the density of the substance.
- ❖ Schottky defect is shown by ionic substances in which the cation and anion are of almost similar size. For example ; (Alkali halides such as) NaCl, KF, KCl, CsCl and AgBr.
- ❖ It may be noted that AgBr shows both Frenkel as well as Schottky defect.



Frenkel Defects

- ❑ Frenkel defect is also called as dislocation defect.
- ❑ This defect is shown by ionic solids.
- ❑ Here the smaller ion usually (cation) is dislocated from its normal site to an interstitial site.
- ❑ It creates a vacancy defect at its original site and an interstitial defect at its new location.
(Combination of vacancy and interstitial atom)
- ❑ It doesn't change the density of the solid.
- ❑ Frenkel defect is shown by ionic substance in which there is a large difference in the size of ions. For example ZnS, AgCl, AgBr and AgI due to small size of Zn and Ag ions.

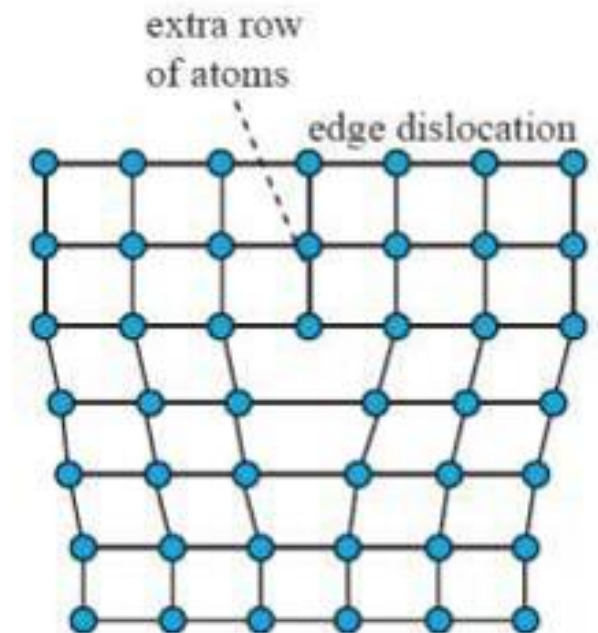
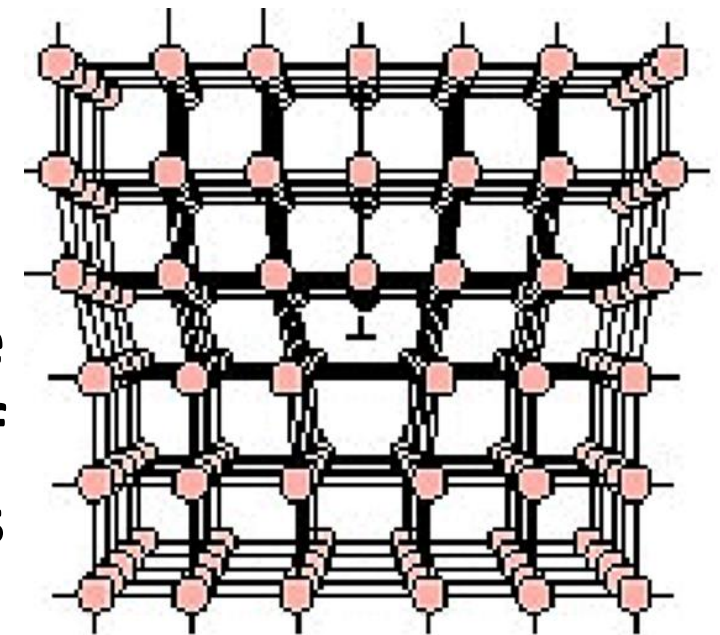


Line Defects

- Defect that is produced due to the misalignment of atoms in a crystal lattice.
- Interatomic bonds significantly distorted in immediate vicinity of dislocation line.
- Dislocation affects the mechanical properties. Line defect produces two types of dislocations;
 - I. Edge Dislocation
 - II. Screw Dislocation
- Dislocation arises due to
 - Growth accidents
 - External stresses causing plastic flow
 - Phase transformations
 - Segregation of solute atoms

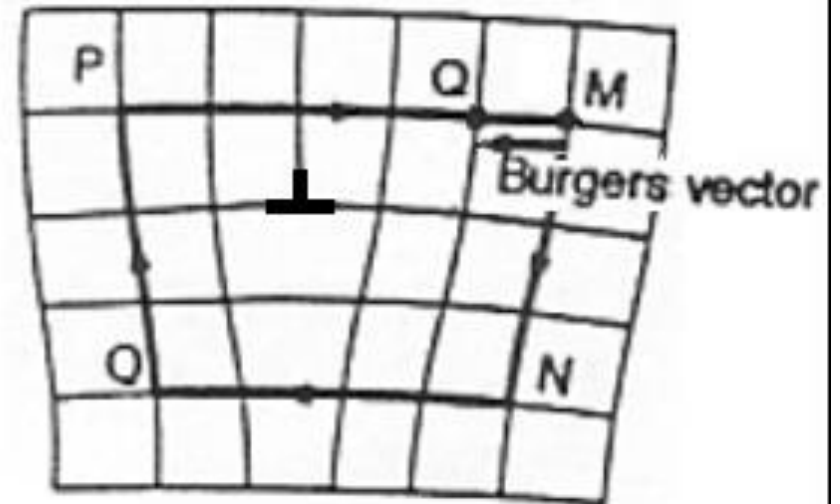
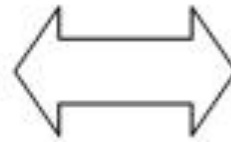
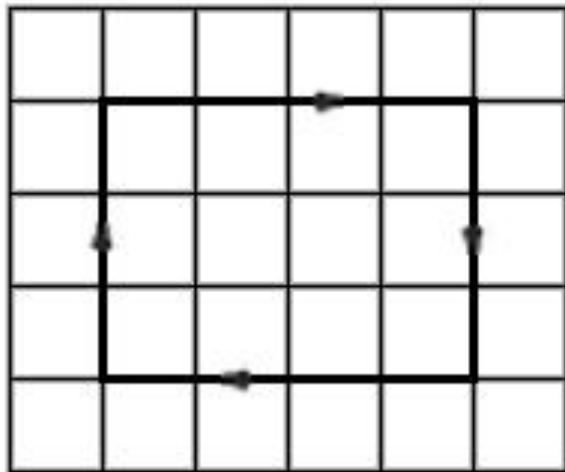
Edge Dislocation

- ❑ It is also called as Taylor-Orowan Dislocation
- ❑ When defect in a crystal occurs due to absence of extra half plane of atoms or presence of half plane of atoms then this type of line defect is called edge dislocation or edge imperfection.
- ❑ The adjacent planes are not straight, but instead bend around the edge of the terminating plane so that the crystal structure is perfectly ordered on either side
- ❑ Edge dislocations make it easier for atoms to slip past one another, making it easier for the metal to deform
- ❑ Two types
 - Positive edge dislocation
 - Negative edge dislocation



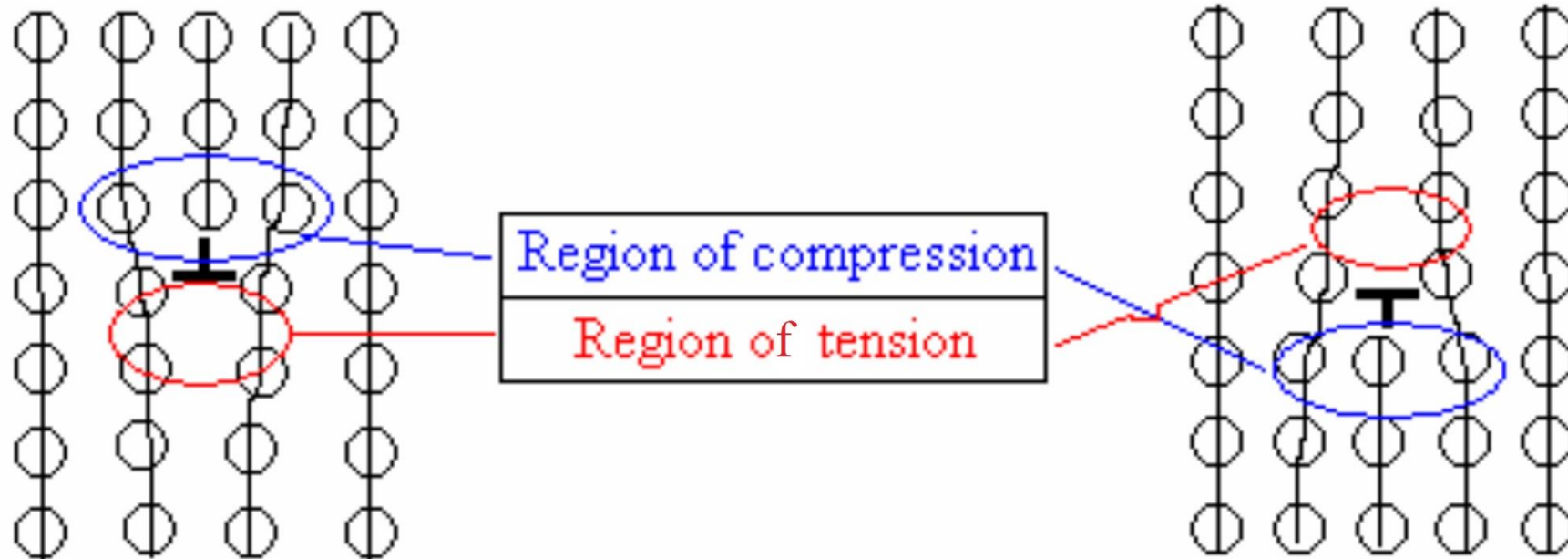
Edge Dislocation

- It may be described as an edge of an extra plane of atoms within a crystal structure.
- Thus regions of compression and tension are associated with an edge dislocation.
- Because of extra incomplete plane of atoms, the atoms above the dislocation line are squeezed together and are in state of compression whereas atoms below are pulled apart and experience tensile stresses.
- Edge dislocation is characterized by a Burger's vector that is perpendicular to the dislocation line.



Edge Dislocation

- Edge dislocation is considered positive when compressive stresses present above the dislocation line, and is represented by $\perp$.
- If the stress state is opposite i.e. compressive stresses exist below the dislocation line, it is considered as negative edge dislocation, and represented by $\top$.
- It is only a convention, because a negative edge dislocation can be converted to a positive edge dislocation by turning over the crystal.

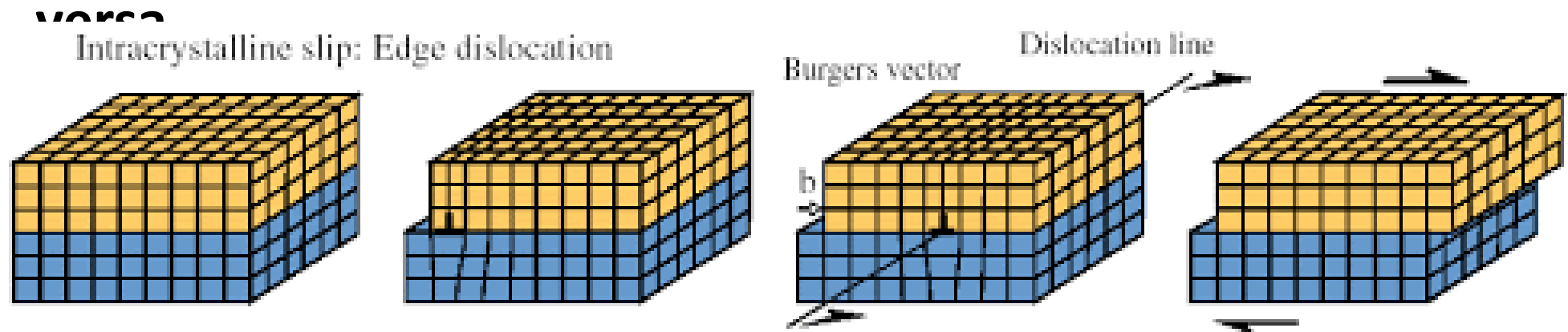


Positive Edge dislocation

Negative Edge dislocation

Edge Dislocation

- A pure edge dislocation can glide or slip in a direction perpendicular to its length i.e. along its Burger's vector in the slip plane, on which dislocation moves by slip while conserving number of atoms in the incomplete plane.
- It may move vertically by a process known as climb, if diffusion of atoms or vacancies can take place at appropriate rate.
- Atoms are added to the incomplete plane for negative climb i.e. the incomplete plane increases in extent downwards, and vice versa



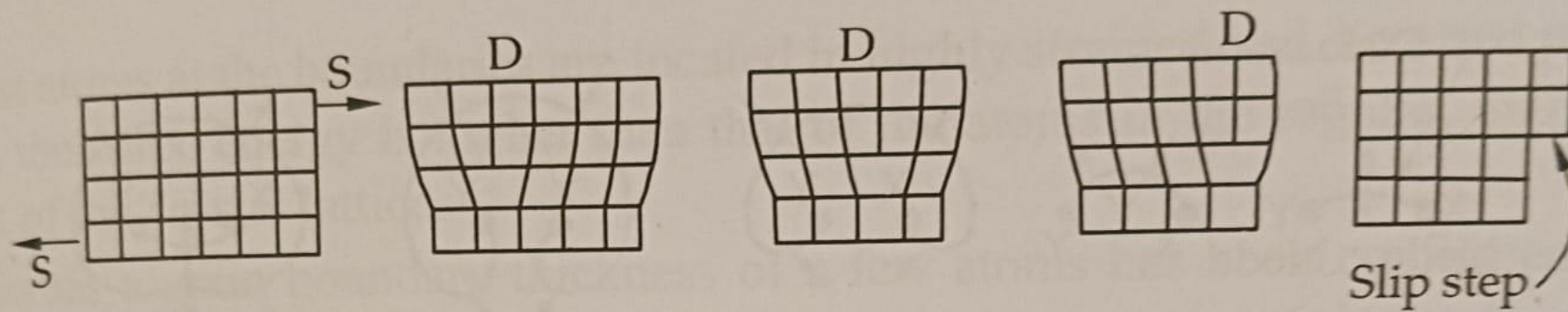
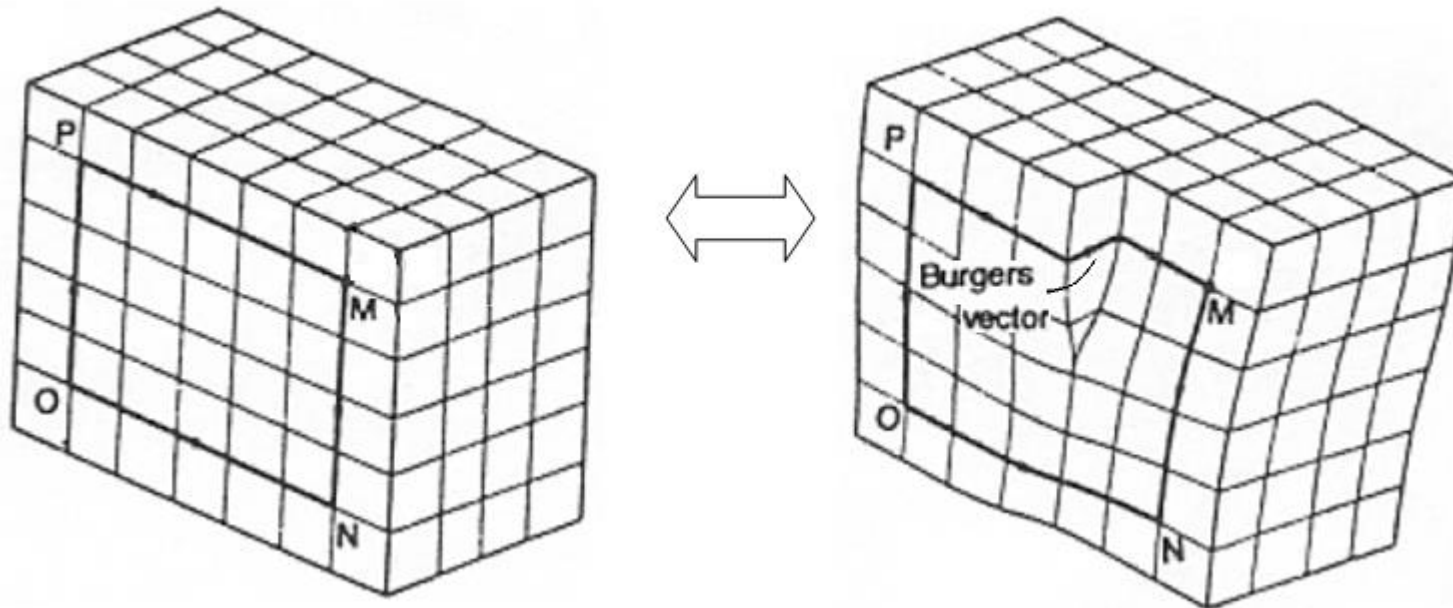
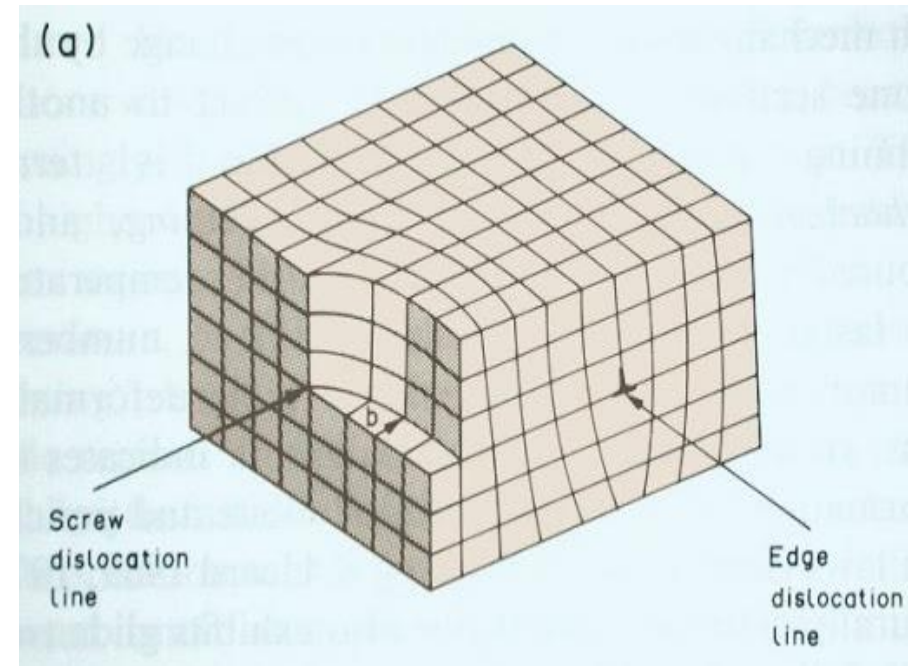


Fig. 37.6 Slip caused by the movement of edge dislocation

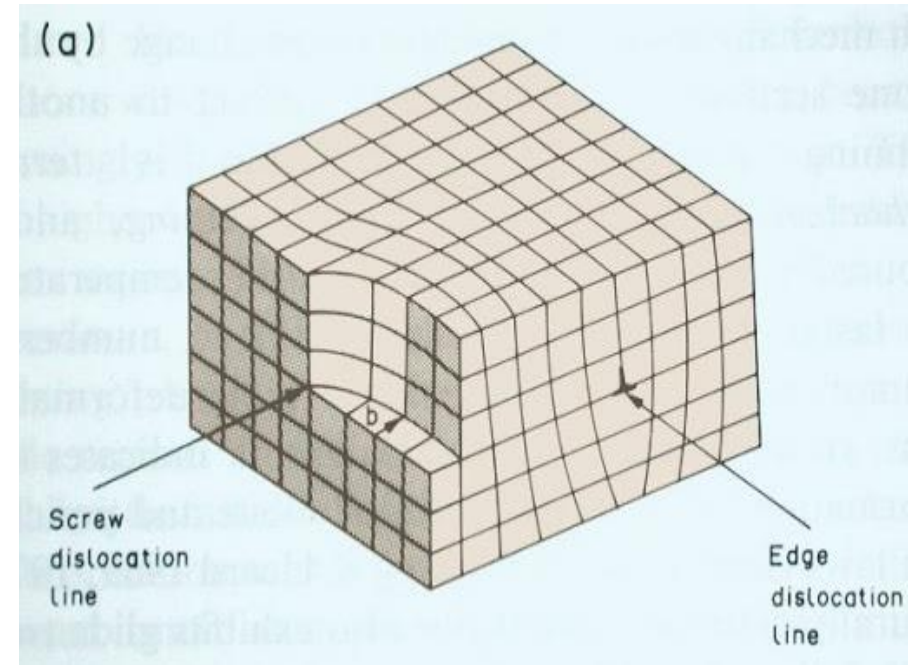
Screw Dislocation

- ❑ It is also called as Burger's Dislocation
- ❑ It is a defect in which the atoms are not arranged in a line but they are in the form of a screw thread which spirals down from one end of the crystal to another in the form of a helix.
- ❑ The lattice/atomic planes follow helical or spiral along the dislocation line.



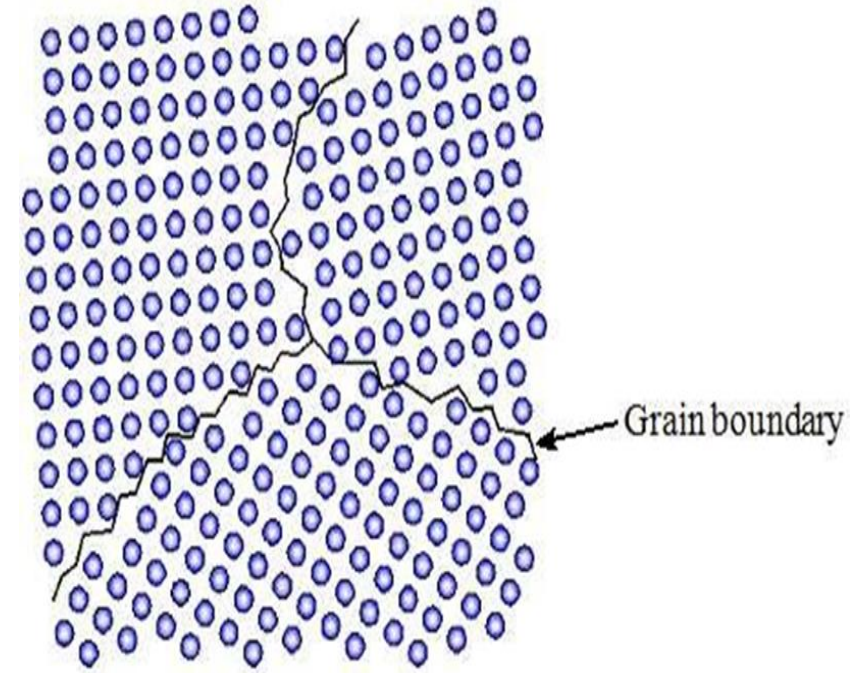
Screw Dislocation

- A screw dislocation can be imagined as being produced by cutting the crystal partway through with a knife and then shearing one part of the crystal with respect to the other parallel to the cut.
- *Burger vector* is parallel to dislocation line.



Planar Or Surface Defects

- Surface defects may arise at the boundary between two grains or small crystals within a layered crystals
- The row of atoms in two different grains may run in slightly different direction, leading to mismatch across the boundary
- Two Dimensional defect.
- Due to change in orientation of the atomic planes and stacking sequence of atomic planes.
- Caused during solidification or mechanical or thermal treatment of material.
- Effect the mechanical properties, electrical resistance and corrosion resistance.



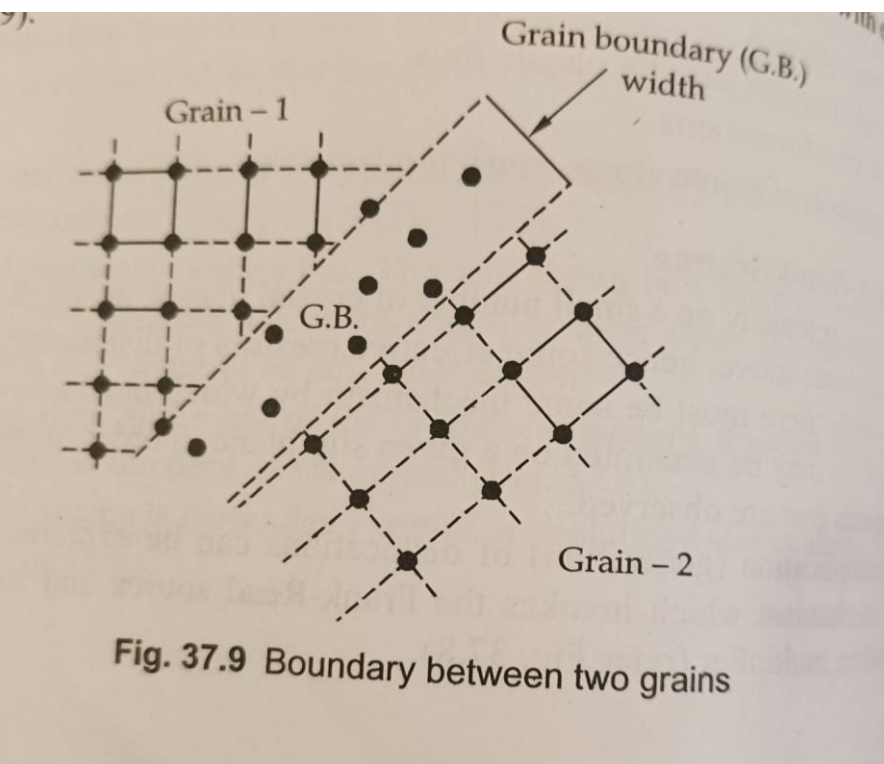
Planar Or Surface Defects

➤ **Planar defects are of following types:**

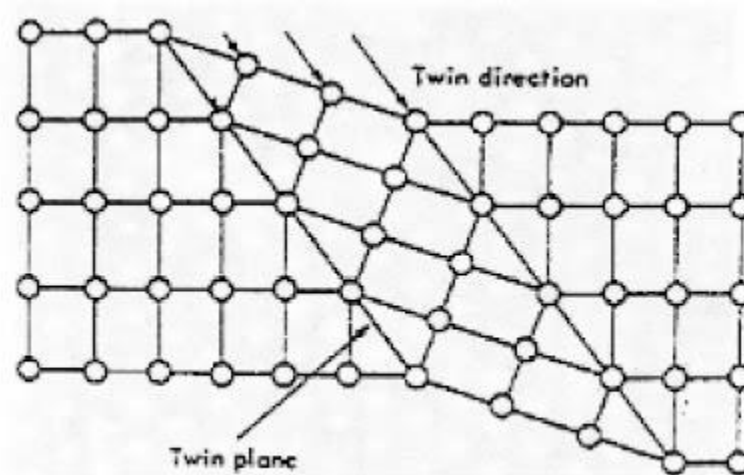
a) Stacking faults

b) Grain boundaries

- **Tilt boundaries**
- **Twin boundaries**
- **Twist boundary**



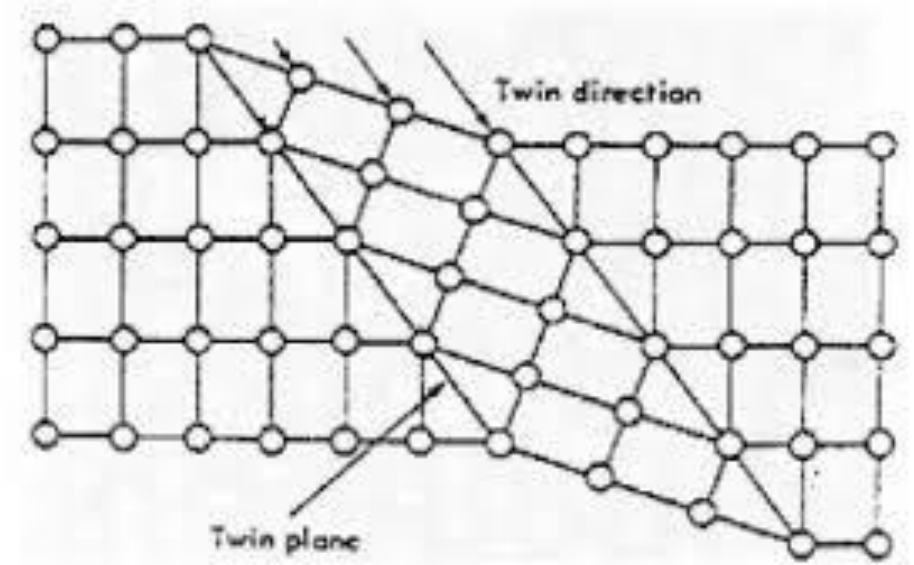
- A twin boundary is a crystal defect that occurs as a planar surface imperfection where two adjacent twin regions meet:
- The atomic arrangement on either side of the twin boundary is a mirror reflection of each other.
- The area between the two boundaries is called the twinned region.
- Twin boundaries occur in pairs, restoring the orientation change caused by the first boundary.



Planar Or Surface Defects

Twin Boundaries

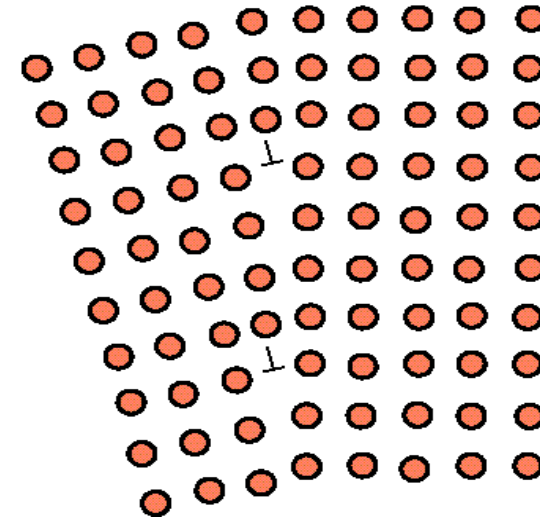
- These are the boundaries in the grains at which the atomic arrangement on one side of the boundary is the mirror image of the atoms on the other side.
- The volume of material which has an orientation similar to the mirror image of the matrix orientation is called a twin.
- May be produced by shear deformation.
- The region between the pair of boundaries is called the twinned region.
- The plane is called twinning plane.



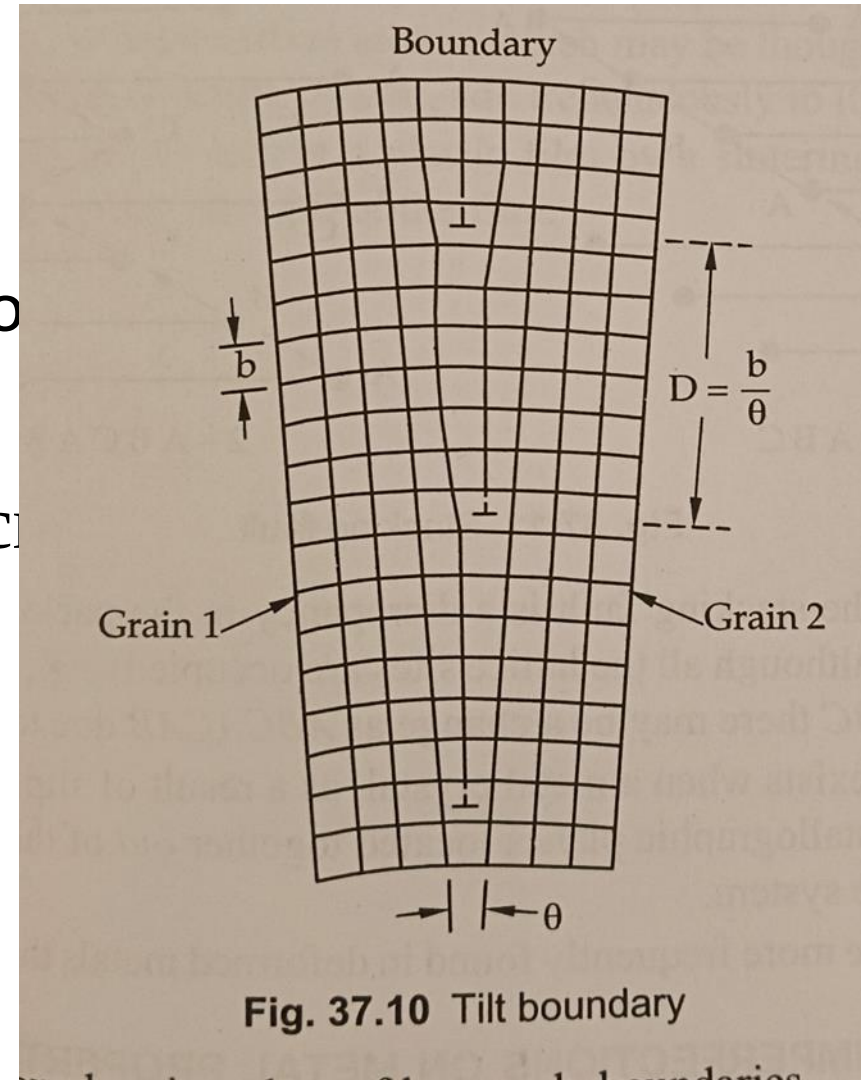
Planar Or Surface Defects

Tilt Boundaries

- When the angle between two crystals is less than 10° , the distortion is not so drastic as to be compared with a non crystalline material. They are also called low angle boundaries.
- It can be described as set of parallel, equally spaced edge dislocation of same sign located one above other.
- A tilt boundary, between two slightly misaligned grains appears as an array of edge dislocations.



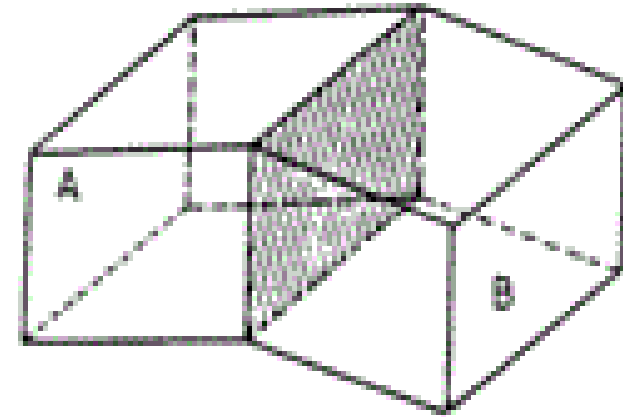
- Tilt boundary is an array of dislocations
- Low angle boundary ($<10^\circ$)
- Twist boundary is an array of screw dislocations (10° to 15°)



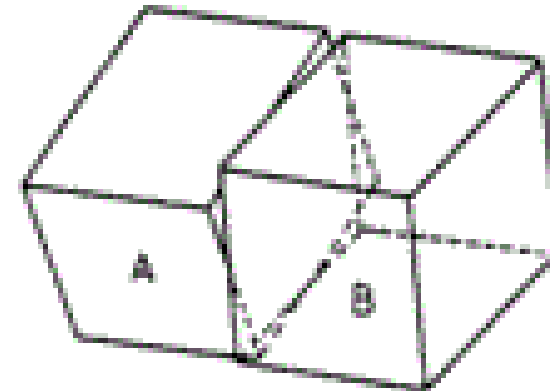
Planar Or Surface Defects

Twist Boundary

- Low angle grain boundaries, that appear as an array of Screw dislocations.



Tilt boundary



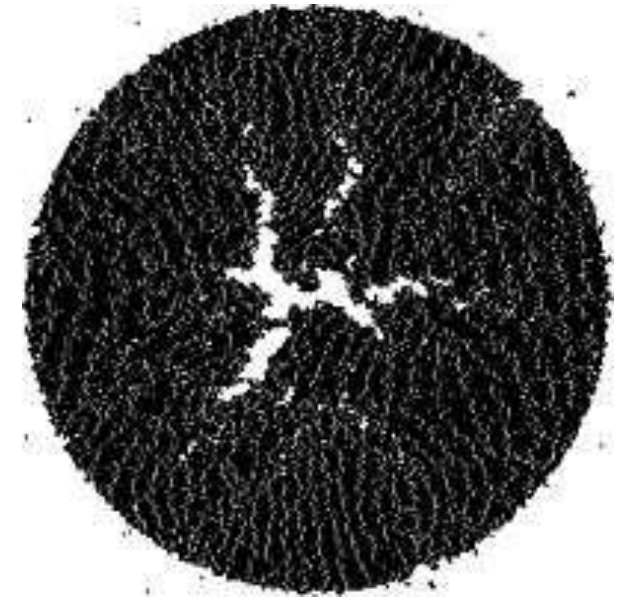
Twist boundary

Volume Defects

- **Volume defects such as stacking faults may arise when there is only small dissimilarity between the stacking sequence of close packed planes in FCC and HCP metals.**
- **Stacking faults are of two types called as intrinsic and extrinsic**
- **Intrinsic fault results in one break whereas extrinsic fault results in two breaks in the sequence.**
- **The volume defects may affect their mechanical, electrical and optical properties.**

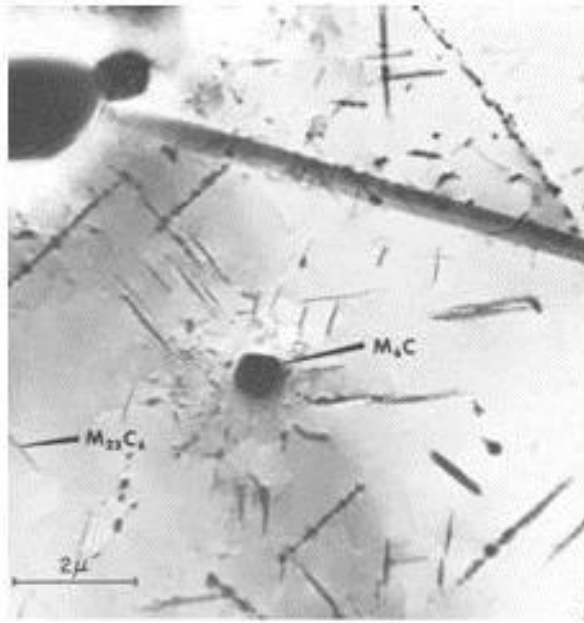
Volume Defects

- ❖ **Precipitates** : Fraction of a micron in size
 - When the reaction occurs in a liquid solution, the solid formed is called the precipitate.
- ❖ **Dispersants** : may be large precipitates, grains, or polygranular particles distributed through microstructure
- ❖ **Inclusions** : foreign particles or large precipitate particles ; undesirable ; harmful
- ❖ **Voids** : Trapped Gases ; Decreases mechanical strength

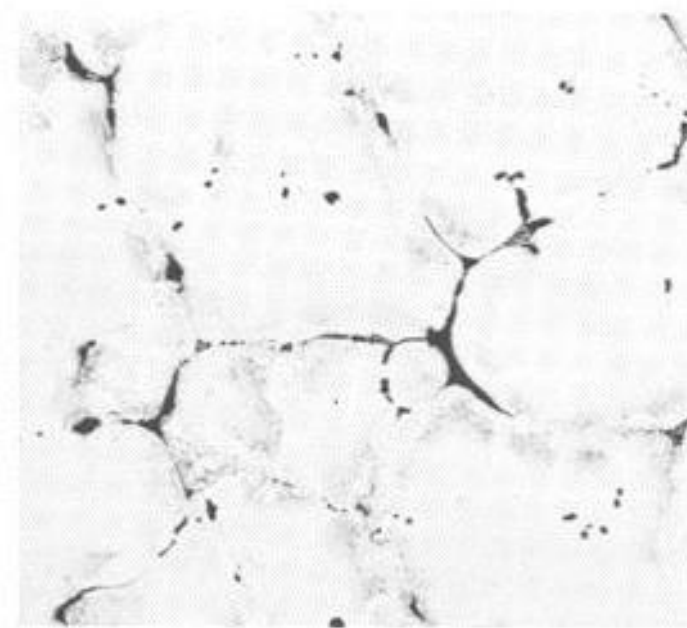


Cluster of microcracks

Volume Defects



Inclusion



Voids/Pores

Angle Between Two Planes

- Calculating the angle between two planes

For cubic crystals, the angle, ϕ between two planes, (h_1, k_1, l_1) and (h_2, k_2, l_2) is given by:

$$\cos \phi = \frac{h_1 h_2 + k_1 k_2 + l_1 l_2}{\sqrt{h_1^2 + k_1^2 + l_1^2} \sqrt{h_2^2 + k_2^2 + l_2^2}}$$

- Example: Calculate the angle between the (111) and (200) planes.

From the above,

$$\cos \phi = \frac{(1 \times 2) + (1 \times 0) + (1 \times 0)}{\sqrt{1+1+1} \sqrt{4+0+0}}$$

$$\cos \phi = \frac{1}{\sqrt{3}}$$

which produces the result, $\phi = 54.75^\circ$

- For two directions to be perpendicular, $\phi = 90^\circ$, $\cos \phi = 0$

ie, $h_1 h_2 + k_1 k_2 + l_1 l_2 = 0$

Equilibrium Concentration Of Vacancies

- The equilibrium number of vacancies N_v for a given quantity of material depends on and increases with temperature according to

$$\frac{N_v}{N} = \exp\left(-\frac{Q_v}{kT}\right)$$

- In this expression, N is the total number of atomic sites, Q_v is the energy required for the formation of a vacancy, T is the absolute temperature in kelvin, and k is the Boltzmann's constant.
- The value of k is 1.38×10^{-23} J/atom-K, or 8.62×10^{-5} eV/atom-K, depending on the units of Q_v .
- Thus, the number of vacancies increases exponentially with temperature; that is, as T in above equation increases, so does also the expression $\exp\left(-\frac{Q_v}{kT}\right)$.
- For most metals, the fraction of vacancies $\frac{N_v}{N}$ just below the melting temperature is on the order of 10^{-4} ; that is, one lattice site out of 10,000 will be empty.

Equilibrium Concentration Of Vacancies

- Calculate the equilibrium number of vacancies per cubic meter for copper at 1000°C. The energy for vacancy formation is 0.9 eV/atom; the atomic weight and density (at 1000°C) for copper are 63.5 g/mol and 8.40 g/cm³, respectively.
- Solution:
- To determine the value of N , the number of atomic sites per cubic meter for copper, from its atomic weight A_{Cu} , its density ρ , and Avogadro's number N_A , according to

$$\begin{aligned} N &= \frac{N_A \rho}{A_{Cu}} \\ &= \frac{(6.023 \times 10^{23} \text{ atoms/mol})(8.40 \text{ g/cm}^3)(10^6 \text{ cm}^3/\text{m}^3)}{63.5 \text{ g/mol}} \\ &= 8.0 \times 10^{28} \text{ atoms/m}^3 \end{aligned}$$

Equilibrium Concentration Of Vacancies

- Thus, the number of vacancies at 1000°C (1273 K) is equal to

$$\frac{N_v}{N} = \exp \left(- \frac{Q_v}{kT} \right)$$

$$N_v = N \exp \left(- \frac{Q_v}{kT} \right)$$

$$= (8.0 \times 10^{28} \text{ atoms}/m^3) \exp \left[- \frac{(0.9 \text{ eV})}{(8.62 \times 10^{-5} \text{ eV/K})(1273 \text{ K})} \right]$$

$$= 2.2 \times 10^{25} \text{ vacancies}/m^3$$

Diffusion in solids

- The movement or jumping of atoms, within the solid is called diffusion.
- As the temperature is increased, the increased energy permits the atoms to vibrate into greater or shorter interatomic distances.
- The activation energy for diffusion (Q) (cal /mole, J/atom or eV/ atom) depends upon the following factor:-
 - 1- Atomic size :Greater the atomic size, higher Q
 - 2- Bond strength :Greater the bond strength of solid higher Q
- Vacancy diffusion mechanism needs higher Q than that for interstitial diffusion mechanism

Classification of diffusion

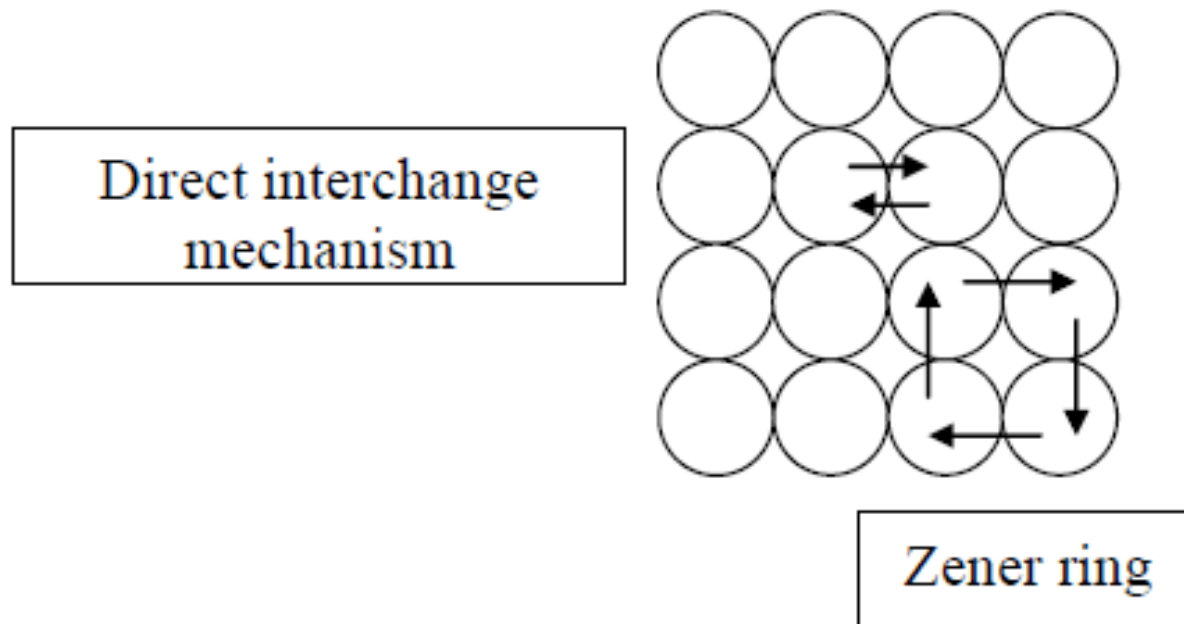
- **1- Self diffusion:-** The movement of atoms through their own structure is called self diffusion. This type studied experimentally by adding a trace of radioactive isotopes of Ni, into a normal Ni.
- **2- Inter diffusion:-** The movement of atoms in two different metals from both sides into each other is called inter diffusion. The inter diffusion is observed in binary metal alloys such as Cu- Ni system.
- **3- Grain boundary diffusion:-** The movement of atoms along the grain boundaries of a material is called grain boundary diffusion.
- **4- Surface diffusion:-** The movement of atoms along the surface of material is called surface diffusion

- Fick's first law gives the relation between the flux (the amount of substance flowing in unite time through unite surface area perpendicular to the flux, and the concentration gradient of the diffusing substance.
- $J = -D (\partial C / \partial X)$
- where:- J= the flux
- C= concentration perpendicular to the direction of flow.
- $(\partial C / \partial X)$ = concentration gradient.
- D= diffusion coefficient and it is a measure of the rate at which the system tends to eliminate the concentration difference (cm²/sec).

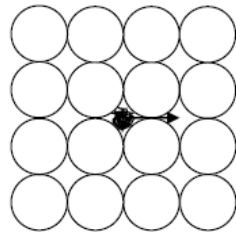
- Fick's second law gives the relation between the concentration gradient and the rate of change of concentration caused by:- diffusion at a point in the system:-
- $\partial C / \partial t = D \partial^2 C / \partial X^2$
- The relationship between (D) and temperature:-
- $D = D_0 \exp -Q / RT$
- Where :- D_0 = frequency factor (cm^2/sec).
- Q = activation energy of diffusion.
- R = gas constant ($1.987 \text{ cal. mol} / ^\circ\text{K}$) or ($8.314 \text{ J} / \text{mol. } ^\circ\text{K}$)
- T = absolute temperature. ($^\circ\text{K}$)

- **Factors affecting coefficients of diffusion:-**
- **1- Temperature:-** (D) increases exponentially with the increase in temperature.
- **2- Crystal structure:-** (D) is higher for those crystal structures which have low packing density.
- **3- Atomic radius:-** (D) in a binary solution is higher if the atomic radii of the solute and solvent different.
- **4- Melting temperature:-** (D) is higher for the solids having lower melting temperatures.
- **5- Grain size:-** (D) is higher for fine grain size materials than coarse grain size materials

- **Diffusion Mechanisms:-**
- **1- Direct Interchange Mechanism.** Two atoms interchange their positions; this process requires a large amount of energy in distorting the crystal as the atoms pass.
- **The Zener Ring mechanism of diffusion avoids the problem of having the atoms squeeze past one another, it involves a ring of atoms that a**
ring of **shift.**



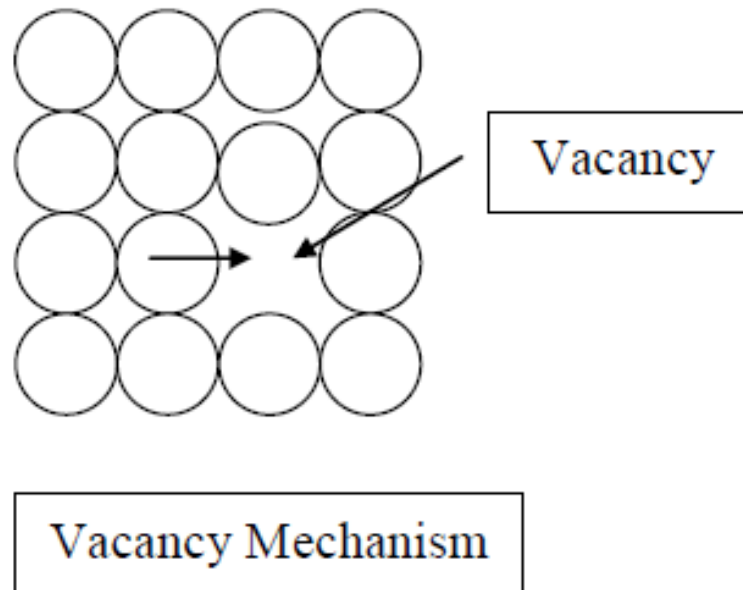
- **2- Interstitial Mechanism.**
- **This type of mechanism occurs in those binary metallic alloys whose atomic radii differ in size significantly**
- **diffusion of carbon atoms in FCC or BCC iron**
- **(atomic radius of C atom is 0.077 nm while that of FCC iron is 0.127 nm).**



Interstitial Mechanism

3- Vacancy Mechanism.

The diffusion occurs because of the atoms jumping into the vacant sites, it may be noted that as the atoms move in one direction through vacancy mechanism the vacancies move in the opposite direction.



Steady-state Diffusion

- Diffusion is a time-dependent process—that is, the quantity of an element that is transported within another is a function of time.
- Often it is necessary to know how fast diffusion occurs, or the rate of mass transfer.
- This rate is frequently expressed as a diffusion flux (J), defined as the mass (or, equivalently, the number of atoms) M diffusing through and perpendicular to a unit cross-sectional area of solid per unit of time.

- In mathematical form, this may be represented as
$$J = \frac{M}{At}$$
where A denotes the area across which diffusion is occurring and t is the elapsed diffusion time.

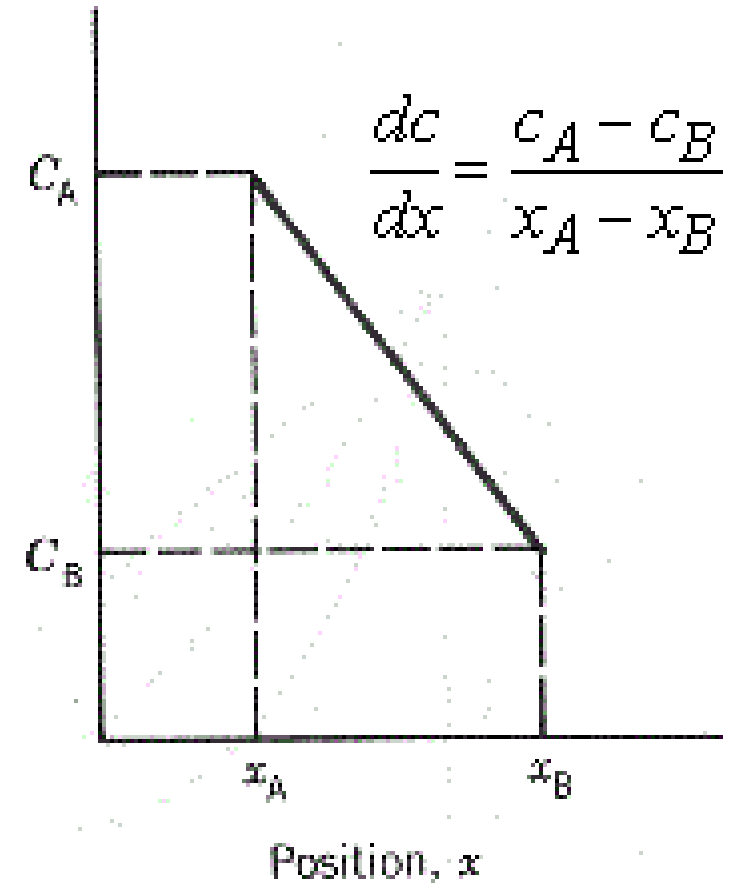
- In differential form, this expression becomes
$$J = \frac{1}{A} \frac{dM}{dt}$$
- The units for J are kilograms or atoms per meter squared per second ($\text{kg}/\text{m}^2\text{-s}$ or $\text{atoms}/\text{m}^2\text{-s}$).
- If the diffusion flux does not change with time, a steady-state condition exists.

Steady-state Diffusion

- When concentration C is plotted versus position (or distance) within the solid x , the resulting curve is termed the concentration profile $[C(x)]$; the slope at a particular point on this curve is the concentration gradient ($\frac{dC}{dx}$).

- Here, the concentration profile is assumed to be linear.

- Therefore, concentration gradient = $\frac{\Delta C}{\Delta x} = \frac{C_A - C_B}{x_A - x_B}$



Steady-state Diffusion

- For diffusion problems, it is sometimes convenient to express concentration in terms of mass of diffusing species per unit volume of solid (kg/m^3 or g/cm^3).
- The mathematics of steady-state diffusion in a single (x) direction is relatively simple, in that the flux is proportional to the concentration gradient through the expression

$$J = -D \frac{dC}{dx}$$

- The constant of proportionality D is called the diffusion coefficient, which is expressed in square meters per second. The negative sign in this expression indicates that the direction of diffusion is down the concentration gradient, from a high to a low concentration.
- This equation is sometimes called **Fick's first law**.
- For diffusion reactions, several driving forces are possible; but when diffusion is according to Fick's first law, the concentration gradient is the driving force.

Steady-state Diffusion

- A plate of iron is exposed to a carburizing (carbon-rich) atmosphere on one side and a decarburizing (carbon-deficient) atmosphere on the other side at 700°C (1300 F). If a condition of steady state is achieved, calculate the diffusion flux of carbon through the plate if the concentrations of carbon at positions of 5 and 10 mm (5×10^{-3} and 10×10^{-3} m) beneath the carburizing surface are 1.2 and 0.8 kg/m³, respectively. Assume a diffusion coefficient of 3×10^{-11} m²/s at this temperature.

- Solution:

Fick's first law is utilized to determine the diffusion flux.

Substitution of the values above into the expression yields

$$\begin{aligned} J &= -D \frac{C_A - C_B}{x_A - x_B} = - (3 \times 10^{-11}) \left(\frac{1.2 - 0.8}{5 \times 10^{-3} - 10 \times 10^{-3}} \right) \\ &= 2.4 \times 10^{-9} \text{ kg/m}^2\text{-s} \end{aligned}$$

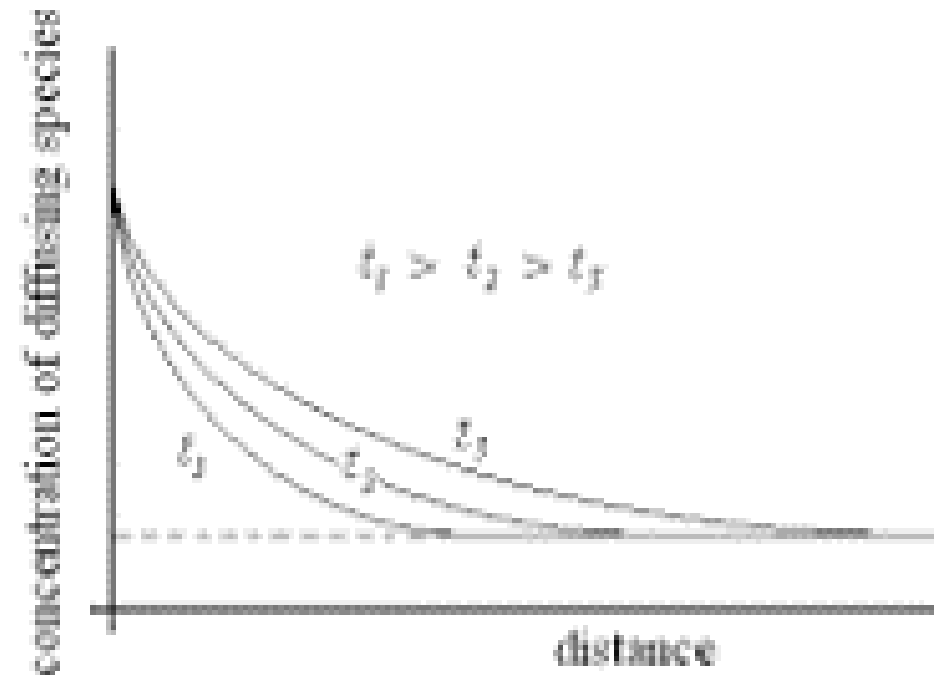
Nonsteady-state Diffusion

- Most practical diffusion situations are nonsteady-state ones.
- That is, the diffusion flux and the concentration gradient at some particular point in a solid vary with time.
- Under conditions of nonsteady state, the partial differential equation is as follows

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right)$$

- This equation is known as Fick's second law.
- If the diffusion coefficient (D) is independent composition, then

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$



Non-steady-state diffusion